



Effect of discreteness on domain wall stability in a plate coupled to a foundation of bistable elements

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ABSTRACT

Surfaces and structures capable of multiple stable configurations have attracted growing interest for on-demand shape morphing. In this work, we consider an elastic compliant plate coupled to a two-dimensional foundation comprising an array of bistable elements, a system that can form and retain complex continuous morphologies without sustained actuation via creation of stable domain walls separating regions in different stable states. These domain walls exhibit three distinct behaviors: expansion, shrinking, and metastable pinning. These arise from two limits of foundation discreteness. In the continuum limit, where bistable units are strongly coupled, domain walls undergo global phase transitions analogous to first-order phase transitions. In the anti-continuum limit, discreteness introduces Peierls-Nabarro type energy modulations that lead to metastable pinning. To quantify these behaviors and the transition between the two limits, we develop a reduced-order model that captures the total potential energy of configurations with domain walls and validate it using finite element analysis (FEA). For axisymmetric domain walls, the model yields phase diagrams identifying the regimes of expansion, shrinking, and pinning as functions of bistable-potential asymmetry, relative foundation discreteness, and domain-wall size. We then extend the analysis to non-axisymmetric geometries and establish local geometric criteria that predict the stability of convex and concave polygonal domain walls, confirmed by simulations. Together, these results clarify how discreteness enables stability through energy-landscape modulation, provide predictive design rules for multistable reconfigurable surfaces, and offer insights into domain-wall stability more generally in elastically coupled multistable metamaterials.

1. Introduction

Surface morphology, characterized by features such as curvature and pattern periodicity, plays a crucial role in governing physical properties across a wide range of length scales and domains, including mechanics (Cerdea and Mahadevan, 2003; Bhushan, 2013), acoustics (Chen et al., 2010; Hussein et al., 2014), optics (Hendrikx et al., 2018; Shen et al., 2021), and fluid dynamics (Barbarino et al., 2011; Peretz et al., 2020). This strong connection between geometric configuration and functional response has motivated growing interest in reconfigurable surfaces and structures that allow on-demand tuning of shape and, consequently, functionality during operation (Barbarino et al., 2011; Dai et al., 2013; Cui and Santer, 2015; Rafsanjani et al., 2019; Yang et al., 2020; Li and

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Pellegrino, 2020; Udani and Arrieta, 2021; Li et al., 2021; Zhang et al., 2022; Liu et al., 2022; Boston et al., 2022; Rahman et al., 2024; Shen et al., 2024; Wu et al., 2025; Liu et al., 2025). While most conventional compliant surfaces require sustained external actuation to maintain a deformed configuration, mechanical bistability (Santer and Pellegrino, 2008; Chi et al., 2022), characterized by non-convex potential energy landscapes typically represented by a double-well profile, enables passive retention of new stable configurations without continuous energy input. Leveraging this principle, recent studies have explored reconfigurable surfaces formed by coupling compliant structures with arrays of bistable elements. In this work, we focus on a system comprising an elastic compliant plate coupled to a two-dimensional array of bistable structural elements. Previous work (Zhang et al., 2022) has experimentally shown that a similar system can realize a wide range of stable, curved surface shapes. Compared to purely discrete reconfigurable architectures, the presence of the plate provides a continuous surface, broadening potential application opportunities. Apart from this application interest, the plate naturally provides long-range elastic interaction between neighboring bistable elements, distinguishing this system from classical nearest-neighbor models such as the ϕ^4 lattice (Kevrekidis and Weinstein, 2000) and from many existing multistable mechanical metamaterial arrays (Nadkarni et al., 2016; Raney et al., 2016; Jin et al., 2020; Zhou and Wang, 2023; Janbaz and Coulais, 2024; Jiao et al., 2024).

The multiple stable configurations in such a system emerge from the collective interplay between the compliance of the plate and the bistability of the foundation units. In addition to the effect of the bistable units on the curvature of the plate, the plate itself causes coupling between adjacent bistable units, and thus the collective behavior is not simply determined by the bistability of isolated units. This could lead to a rich set of possible nonlinear effects. When the coupling between adjacent units is relatively weak, individual elements behave nearly independently (Zhang et al., 2022) (as also shown in other systems, e.g., Refs. Chirikjian, 1994; Santer and Pellegrino, 2008; Cui and Santer, 2015; Udani and Arrieta, 2021; Tahidul Haque et al., 2024), allowing the array to access a large number of metastable configurations. In the anti-continuum limit, where coupling is negligible compared to the unit onsite stiffness, local actuation induces only local transitions, and the number of stable states scales as 2^N (Chirikjian, 1994), with N the number of units. At the opposite extreme (continuum limit), the neighboring elements are strongly elastically coupled, and a localized snap-through induces a dynamic dissipative or diffusive global phase transition with the domain wall which separates two different stable regions propagating through the whole system (Nadkarni et al., 2016; Raney et al., 2016; Jin et al., 2020; Zhou and Wang, 2023; Janbaz and Coulais, 2024; Jiao et al., 2024). This collective behavior is analogous to a first-order phase transition (Oxtoby, 1998; Binder, 1987; Jiao et al., 2024), in which the entire medium converts from one phase to the other. Due to the spontaneous moving domain wall, such systems remain effectively bistable rather than multistable. In this work, we are interested in understanding the transition between these two extremes. Characterizing this transition regime is essential for the rational design of multistable reconfigurable surfaces and for enabling controlled shifts between bistability and multistability, yet systematic studies of this regime in such a system remain limited.

The main goals of this work are twofold. First, we study the transition from strongly-coupled global phase transition behavior (continuum limit) to weakly-coupled multistable behavior (anti-continuum limit) in the plate-bistable foundation system, and we establish criteria for the system parameters necessary to achieve multistability. Inspired by the similarity between the domain-wall motion in this system and the dislocation or phase boundary motion in crystals, we find that the multistability of the domain wall originates from the relative discreteness of the foundation, which leads to periodic potential energy modulation and metastable domain-wall pinning analogous to the Peierls-Nabarro (P-N) effect (Kevrekidis and Weinstein, 2000; Truskinovsky and Vainchtein, 2003). The discreteness can be quantified by the ratio between the lattice spacing of the bistable foundation units and the characteristic width of the domain wall. This discreteness ratio reflects the competition between the bistability of individual foundation units and the cooperative interactions mediated by the plate. To quantitatively assess how discreteness reshapes the energy landscape, we use collective coordinate approximation (Gervais and Sakita, 1975; Takyi and Weigel, 2016) and develop an analytical reduced-order model (ROM) that applies consistently in both the continuum and discrete regimes. Second, within the multistable regime, we identify which domain-wall geometries remain stable. We begin with simple axisymmetric domain walls. Our model shows that metastability arises only within a certain domain-wall size range, reflecting the competition between discreteness effects (P-N pinning) and the global energetic drive for phase transition. We then establish a local-global stability criterion to infer the stability of general polygonal domain walls. Both the axisymmetric and polygonal domain-wall results from our model are validated against Finite-Element Simulation (FEA).

The remainder of the paper is organized as follows. Section 2 introduces the plate-bistable foundation system and illustrates three representative domain-wall behaviors: expansion, shrinking, and metastable pinning. It also provides a qualitative explanation of the underlying mechanisms that give rise to these behaviors. Section 3 develops the reduced-order model (ROM) and derives the potential energy for both the continuous and discrete foundation cases. Section 4 employs the ROM to quantify the critical nucleation radius r_{cr} , which separates expansion from shrinking, and to analyze domain-wall dynamics in the continuous regime, and the effect of discreteness on stability in the discrete regime. Section 5 extends the stability analysis to general polygonal domain walls. Section 6 concludes with a summary and discussion.

2. Expansion, shrinking, or metastable pinning of a domain wall

We consider an elastic thin plate with thickness h , bending stiffness $D = \frac{Eh^3}{12(1-\nu^2)}$, and density ρ , connected to the ground by an elastic foundation consisting of a two-dimensional array of bistable units, as shown in Fig. 1(a). To ensure isotropy, the bistable units are arranged in a hexagonal honeycomb pattern with an edge length d . In this work, we do not specify a particular geometric design for the bistable unit, since bistability can be realized through a variety of mechanisms. Instead, we adopt a general quartic potential

with tunable asymmetry parameter α to represent the energy landscape of an individual bistable unit in a design-independent manner:

$$U_{\text{unit}}(w) = k_{\text{unit}} \left[\frac{1}{4} w^4 - \frac{1}{3} (1 + \alpha) w_b w^3 + \frac{1}{2} \alpha w_b^2 w^2 \right] \quad (1)$$

where k_{unit} is the stiffness, w is the out-of-plane displacement of the bistable unit, w_b is the displacement from the initial stable equilibrium state O to the another stable equilibrium state B , and α characterizes the asymmetry of the bistable potential. For $\alpha > 0.5$, $U_{\text{unit}}(w = w_o = 0) < U_{\text{unit}}(w = w_b)$ and vice versa, as illustrated in Fig. 1(b). According to the geometry of the foundation pattern the effective stiffness per unit area for the foundation is defined as $k = \frac{4}{3\sqrt{3}} k_{\text{unit}} / d^2$.

Now we consider a scenario where a circular region of the foundation is imposed (actuated) via displacement control to deform from the initial state O to the other stable state B , as shown in Fig. 1(c). Due to the elastic coupling from the plate, a domain wall with radius r_{d0} forms, separating the interior region B and the exterior region O. Within the domain wall, the bistable units are in an intermediate position between state O and B . The domain wall, with a width W , can span multiple units. We define the discreteness parameter γ by $\gamma = d/W$ to reflect the relative density of the bistable units inside the domain wall. Next, we release all the displacement boundary condition, and consider how the initial deformation of the plate (effectively the domain wall) evolves.

We first examined the evolution of the domain wall using finite element simulations in *Abaqus*. The plate was modeled as a hexagon and discretized using a structured triangular mesh of S3R shell elements (three-node linear triangular shell elements). A mesh-convergence study was performed to ensure that the mesh size was sufficiently small relative to the domain-wall width. The hexagonal plate shape and triangular mesh were chosen primarily for geometric compatibility with the underlying hexagonal pattern of the bistable foundation. Each foundation unit was represented by a SPRING1 element acting in the out-of-plane (z) direction. In *Abaqus*, the SPRING1 element is a massless spring connecting a plate node to the ground, and its nonlinear force–displacement relation can be prescribed directly in the `.inp` file. Here, the bistable potential in Eq. (1) and the corresponding force–displacement law were implemented through the tabulated nonlinear spring response. All the edges of the hexagonal plate were left free. In the finite element simulations, we focus on domain walls whose size is sufficiently smaller than the overall plate size that the wall remains well separated from the plate boundary. Under this condition, the boundary effect can be neglected. To maintain consistency and enforce nondimensionality, all model parameters (h, D, w_b) were set to unity, except for the unit spacing d and the bistable-element stiffness k_{unit} . For each simulation case, d and k_{unit} were jointly varied to maintain a constant effective foundation stiffness, $k = \frac{4}{3\sqrt{3}} \frac{k_{\text{unit}}}{d^2} = 1$,

thus ensuring that the characteristic length $L = \sqrt[4]{\frac{D}{kw_b^2}}$ of the system (see Section 3) is unity and that the domain-wall width W remains fixed (see Section 3 for the normalization and discussion of the domain-wall width). Under this construction, the discreteness parameter $\gamma = d/W$ becomes directly proportional to d .

The simulation procedure consisted of two sequential steps, both conducted using implicit dynamic analysis. In the first step, a prescribed out-of-plane displacement was applied over a circular region of radius r_{d0} on the plate to nucleate a domain wall. To reduce inertial effects during loading, the prescribed displacement was ramped linearly from 0 to 1 (normalized by w_b) over a time interval of $20T$, where T is the characteristic time scale (see Section 3). In the second step, this displacement constraint was removed and the structure was allowed to evolve freely. To enable the system to relax toward nearby stable equilibria, Rayleigh damping was introduced to dissipate transient dynamics.

Three representative domain-wall evolution behaviors are presented in Fig. 1(d). The evolution of a domain wall is governed by the system's potential energy V , which includes contributions from both the bending energy of the plate and the bistable elements. The bending energy of the plate is localized within the domain wall, which scales approximately (in the first-order) with the perimeter $\sim r_d$ and is referred to as surface energy. The contribution from the bistable elements relates to the energy difference of the transition from states O to B , which approximately scales with r_d^2 and is referred to as the bulk energy. For $\alpha > 0.5$, as the domain wall expands, both the surface energy and the bulk energy increases, giving $V \sim O(r_d) + O(r_d^2)$. The domain wall therefore shrinks after the initial displacement is released. For $\alpha < 0.5$, as the domain wall expands, the surface energy increases but the bulk energy decreases, giving $V \sim O(r_d) - O(r_d^2)$. Competition between the two terms produces a critical radius r_{cr} , below which the $O(r_d)$ term (surface energy) dominates and the domain wall is prone to shrink and above which the $-O(r_d^2)$ term (bulk energy) governs and the domain wall is prone to expand. These two typical behaviors for $\alpha < 0.5$ are observed in FEA simulations in Fig. 1(di) and (diii).

However, Fig. 1(dii) and (div) show that when $\gamma = d/W$ is large enough, the domain wall can stabilize at a finite radius rather than expand or shrink. These metastable states correspond to local minima in the potential energy V , arising from the discreteness of the bistable foundation, an effect analogous to the P-N effect (Kevrekidis and Weinstein, 2000; Truskinovsky and Vainchtein, 2003). The P-N effect in our system can be illustrated by Fig. 2, which conceptually shows the spatial distribution of foundation energy along the radial direction as the domain wall shifts. Considering a process where the domain wall moves from (i) to (ii) and then to (iii). When the foundation lattice size is negligible relative to the domain wall width (γ is small), as in Fig. 2(a), the energy distribution in the foundation remains nearly unchanged throughout the domain wall motion. However, when the lattice size is comparable to the domain wall width (γ is large), as in Fig. 2(b), the energy distributions at positions (i) and (iii) are similar due to lattice periodicity, but differ significantly at the intermediate position (ii) due to lattice discreteness. This results in a periodic variation in V as the domain wall traverses the lattice, giving rise to local minima, i.e., the metastable states.

Although the initial loading area is circular, the final metastable domain wall conforms to the hexagonal symmetry of the foundation. This occurs because the system is more anisotropic at larger γ , and the domain wall naturally evolves toward a shape that minimizes energy with this anisotropy. An analogy can be found in crystal growth: when interfacial energy is anisotropic (as in snowflakes or hexagonal crystals), the equilibrium morphology forms facets rather than circles.

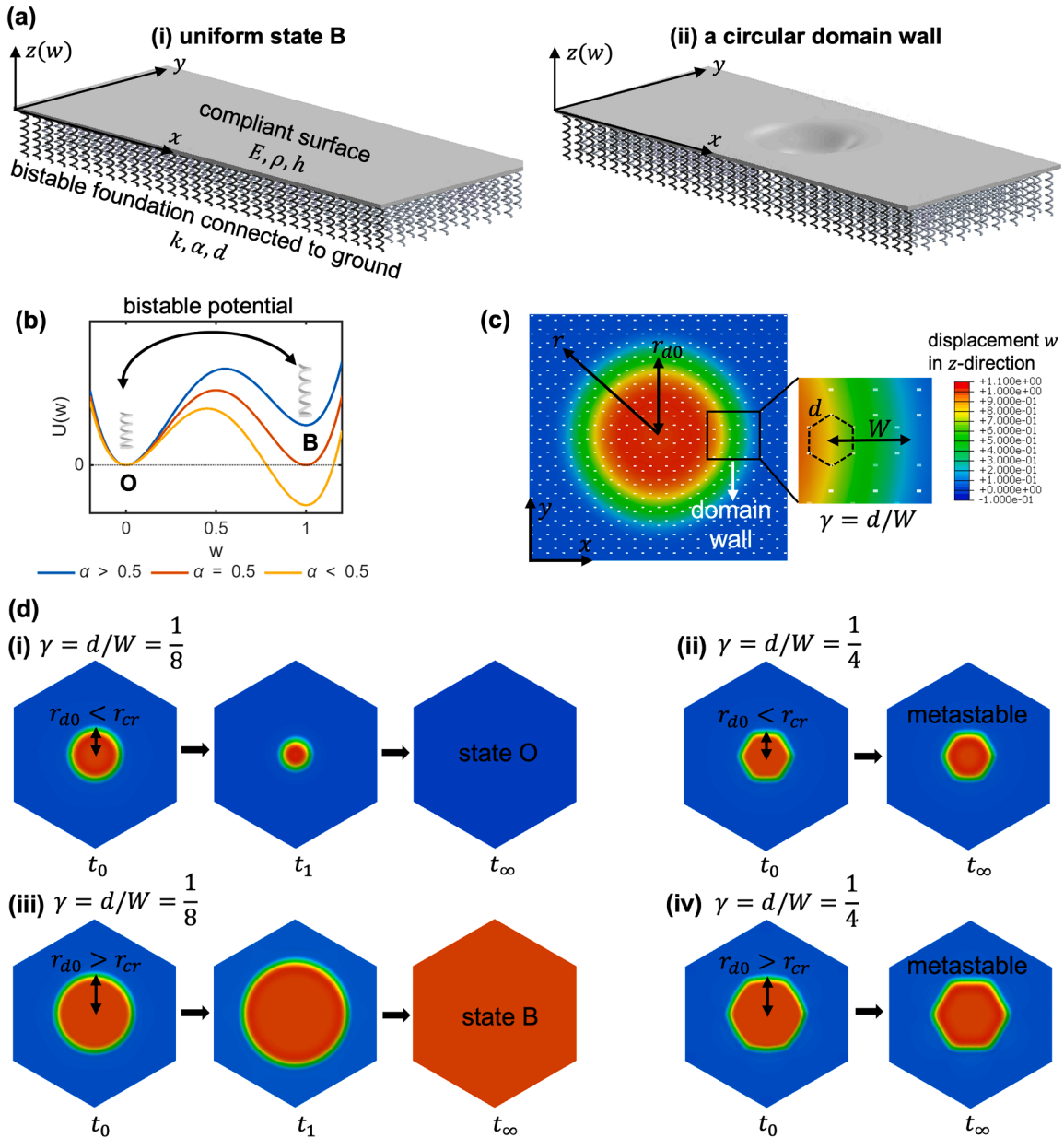


Fig. 1. (a) A compliant and elastic plate is connected to the ground by an elastic bistable foundation: (i) a uniform configuration and (ii) with a circular domain wall. (b) Potential energy landscape $U(w)$ of a bistable foundation unit for different values of α , showing the two stable states O (with displacement in the z direction $w = 0$) and B ($w = w_b$). (c) Displacement boundary condition (normal to the figure graphic) is applied to a circular region, causing the interior region to deform from the original state O to the other stable state B and form a domain wall (in green) with radius r_{d0} and width W . The inset shows the edge size of the hexagonal pattern d . The relative density of the foundation is defined as $\gamma = d/W$. (d) FEA results: (i) and (iii) the domain wall moves when the foundation density γ is relatively small ($\gamma = 1/8$). It shrinks when the initial domain wall size r_{d0} is smaller than a critical size r_{cr} and expands when $r_{d0} > r_{cr}$; (ii) and (iv) the domain wall stabilizes and stays near its initial radius when the foundation density is relatively large with $\gamma = 1/4$. All the FEA results in this plot are with parameters: $h = 1, D = 1, \rho = 0.1, k = 1, w_b = 1, \alpha = 0.48$ and the corresponding characteristic length is $L = 1$. All the dimensions shown here are dimensionless, normalized by L if in the $x - y$ plane and by w_b in the z direction. Moderate damping is used to obtain the metastable configurations.

3. Theoretical model

When the discreteness parameter $\gamma = d/W$ is sufficiently small, the force exerted on the plate by the foundation can be treated as continuously distributed force. We follow the classical Kirchhoff-Love theory for thin plates (Timoshenko and Woinowsky-Krieger,

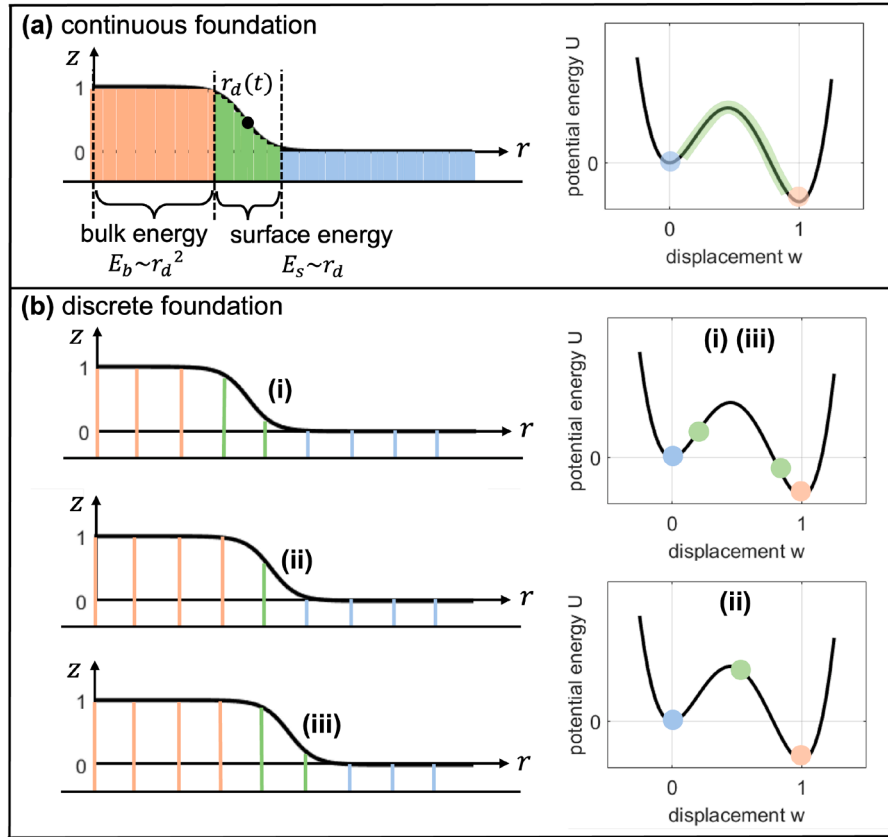


Fig. 2. Conceptual schematic of the Peierls–Nabarro type periodic energy modulation in the bistable foundation: (a) energy distribution for small γ , corresponding to the continuous-foundation limit; (b) energy distribution for large γ , corresponding to the discrete-foundation case. Red denotes state O, blue denotes state B, and green denotes the domain-wall region (the intermediate state between O and B).

1959) and the governing equation for the axisymmetric deformation $w(r, t)$ of the plate-bistable foundation system is :

$$\rho h \frac{\partial^2 w}{\partial t^2} + D \left(\frac{\partial^4 w}{\partial r^4} + \frac{2}{r} \frac{\partial^3 w}{\partial r^3} - \frac{1}{r^2} \frac{\partial^2 w}{\partial r^2} + \frac{1}{r^3} \frac{\partial w}{\partial r} \right) + kw(w - \alpha w_b)(w - w_b) = 0 \quad (2)$$

where r is the radial coordinate with its origin at the domain wall center, $k = \frac{4}{3\sqrt{3}} k_{\text{unit}}/d^2$ is the effective foundation stiffness as defined previously, and the last term is derived from Eq. (1) by taking the first derivative with respect to w . Based on this PDE, we define the characteristic time T and length L ,

$$T = \sqrt{\frac{\rho h}{kw_b^2}}, \quad L = \sqrt[4]{\frac{D}{kw_b^2}} \quad (3)$$

and nondimensional variables:

$$w = w_b \bar{w}, \quad t = T \bar{t}, \quad r = L \bar{r}, \quad (4)$$

leading to the normalized PDE,

$$\frac{\partial^2 \bar{w}}{\partial \bar{t}^2} + \left(\frac{\partial^4 \bar{w}}{\partial \bar{r}^4} + \frac{2}{\bar{r}} \frac{\partial^3 \bar{w}}{\partial \bar{r}^3} - \frac{1}{\bar{r}^2} \frac{\partial^2 \bar{w}}{\partial \bar{r}^2} + \frac{1}{\bar{r}^3} \frac{\partial \bar{w}}{\partial \bar{r}} \right) + \bar{w}(\bar{w} - \alpha)(\bar{w} - 1) = 0. \quad (5)$$

In the present study, the bistable foundation is modeled only through its local bistable restoring force, while its inertia is neglected; accordingly, the model is most accurate when the areal mass density of the foundation is small compared with that of the plate. In the simplest extension, if each foundation unit is kinematically constrained to follow the local out-of-plane motion of the plate and has no independent internal motion, their main effect is to increase the effective inertia of the system, or equivalently its effective mass per unit area, thereby increasing the characteristic time scale and slowing down the domain wall motion. More generally, if the bistable units possess internal dynamic degrees of freedom, they may introduce additional time scales and dynamic coupling with the plate, mainly affecting the transient evolution of the domain wall.

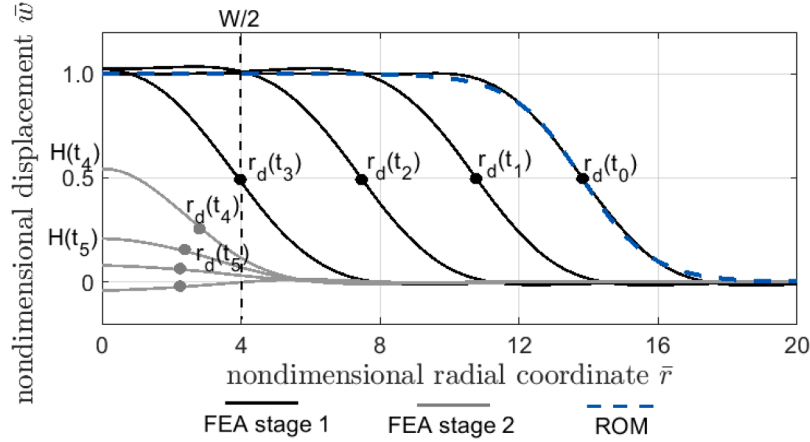


Fig. 3. Two-stage shrinking process of a domain wall. Dots mark the domain wall center positions. FEA results are for $\alpha = 0.48$. Nondimensional out-of-plane displacement $\bar{w} = w/w_b$ and nondimensional radial coordinate $\bar{r} = r/L$ are normalized by w_b and characteristic length L , respectively.

3.1. Reduced-order model

Eq. (5) is closely related to the classical ϕ^4 model, a nonlinear wave equation with a symmetric quartic bistable potential that has been extensively studied over the past several decades (Kevrekidis and Weinstein, 2000). However, unlike the standard ϕ^4 equation, Eq. (5) incorporates a biharmonic operator arising from plate bending, which substantially complicates the analysis and has received only limited analytical treatment to date (Decker et al., 2021). As a consequence, no closed-form solutions of Eq. (5) are available. To overcome this difficulty, we adopt a reduced-order model (ROM) based on a hyperbolic tangent ansatz, Eq. (6), which is motivated by finite-element simulations (Fig. 3) as well as by the known kink solutions of the classical ϕ^4 model,

$$w(r, t) = \frac{1}{2} H \left[1 - \tanh \left(\frac{r - r_d}{W/4} \right) \right], \quad (6)$$

where r_d is the domain wall center position, H the height, and W the width. This general form Eq. (6) captures the plate shapes in two stages of evolution shown in Fig. 3.

Stage 1 ($r_d(t) > W/2$): The domain wall either expands or shrinks (black lines in Fig. 3), with height $H = w_b$. Although W/L in principle depends on the PDE parameter α and the instantaneous position $r_d(t)$, FEA results indicate that this dependence is weak and it can be approximated as a constant $W/L \approx 8$. The factor $W/4$ in the tanh argument ensures that for $r \leq r_d(t) - W/2$, $w(r, t) \approx w_b$ (interior state B), and for $r \geq r_d(t) + W/2$, $w(r, t) \approx 0$ (exterior state O). The plate profile Eq. (6) reduces to:

$$w(r, t) = \frac{1}{2} w_b \left[1 - \tanh \left(\frac{r - r_d(t)}{W/4} \right) \right] \quad (7)$$

Stage 2 ($r_d(t) < W/2$): The domain wall is vanishing (gray lines in Fig. 3). Both the height $H(t)$ and width $W(t)$ vary with time. Since $r_d(t) = W(t)/2$, the two independent, time-dependent parameters are $H(t)$ and $r_d(t)$ (or equivalently $W(t)$):

$$w(r, t) = \frac{1}{2} H(t) \left[1 - \tanh \left(\frac{r - r_d(t)}{r_d(t)/2} \right) \right]. \quad (8)$$

To study the stability and dynamics of the domain wall, we apply the collective coordinate method (Gervais and Sakita, 1975; Takyi and Weigel, 2016). This approach assumes that the displacement field maintains a fixed form (here, a tanh profile), while the system's evolution is governed by a small set of time-dependent parameters (the collective coordinates). In our approximation, Stage 1 is characterized solely by $r_d(t)$, whereas Stage 2 involves both $r_d(t)$ and $H(t)$. This reduced-order model (ROM) enables explicit derivation of an effective potential energy landscape $U(r_d, H)$, from which key mechanical features, such as energy maxima and minima, can be identified. The ROM thus provides the foundation for quantitatively predicting domain wall evolution, stability, and reconfigurability in the plate-bistable foundation system.

3.2. Potential energy for continuous foundation case

We first consider the foundation in a continuous limit where $\gamma = d/W \ll 1$. In stage 1 where the domain wall either shrinks or expands, the elastic energy in the plate (Timoshenko and Woinowsky-Krieger, 1959) is given by

$$\begin{aligned} E_{p1} &= \frac{1}{2} D \iint \left[\frac{1}{r^2} \left(\frac{\partial w}{\partial r} \right)^2 + \left(\frac{\partial^2 w}{\partial r^2} \right)^2 \right] dA \\ &\approx \pi D w_b^2 \int_{-\infty}^{\infty} \left[\frac{4}{W^2} \text{sech}^4(\xi) \frac{1}{r_d} + \frac{16^2}{W^4} \text{sech}^4(\xi) \tanh^2(\xi) r_d \right] \frac{W}{4} d\xi \end{aligned}$$

$$= \frac{\pi D w_b^2}{15 W^2} \left(256 \frac{r_d}{W} + 20 \frac{W}{r_d} \right) \quad (9)$$

where the deformation Eq. (7) is used and Poisson ratio is taken as zero for simplicity without losing the key insights by the ROM.

The elastic energy stored in the foundation is calculated as

$$\begin{aligned} E_{f1} &= \int_0^\infty 2\pi r U(w(r)) dr \\ &= 2\pi k \int_0^\infty \left[\frac{1}{4} w^4 - \frac{1}{3} (1 + \alpha) w_b w^3 + \frac{1}{2} \alpha w_b^2 w^2 \right] r dr \\ &\approx \frac{\pi k w_b^4 W^2}{384} \left[(2\alpha - 1) + 4 \frac{r_d}{W} + 32(2\alpha - 1) \left(\frac{r_d}{W} \right)^2 \right] \end{aligned} \quad (10)$$

where $U(w)$ is the bistable potential Eq. (1) and the ansatz Eq. (7) is used. To yield the above explicit expression, the change of variable $\frac{r-r_d}{W/4} = \xi$ was used and the integration limit $-\frac{r_d}{W/4}$ was well approximated by $-\infty$ due to the strong localization of sech function. The derivation details are presented in Appendix A.

Next, we consider stage 2 where the domain wall disappears. The elastic energy in the plate in stage 2 follows from Eq. (9) with w_b replaced by $H(t)$ and $W(t) = 2r_d(t)$:

$$E_{p2} = \frac{14\pi D}{5} \frac{H^2(t)}{r_d(t)^2} \quad (11)$$

The elastic energy in the foundation is:

$$\begin{aligned} E_{f2} &= \int_0^\infty 2\pi r U(w(r)) dr \\ &= 2\pi k \int_0^\infty \left[\frac{1}{4} w^4 - \frac{1}{3} (1 + \alpha) w_b w^3 + \frac{1}{2} \alpha w_b^2 w^2 \right] r dr \\ &\approx \frac{\pi k w_b^4}{96} \left[11 \left(\frac{H(t)}{w_b} \right)^4 - (18\alpha + 18) \left(\frac{H(t)}{w_b} \right)^3 + 36\alpha \left(\frac{H(t)}{w_b} \right)^2 \right] r_d(t)^2 \end{aligned} \quad (12)$$

which recovers Eq. (10) for $H = w_b$ and $r_d = W/2$.

Defining the characteristic energy based on the characteristic scales as

$$E = k w_b^4 L^2, \quad (13)$$

the energies in Eqs. (9)–(12) can be normalized. For $r_d(t) > W/2 = 4$, we obtain

$$\bar{E}_{p1} = \frac{E_{p1}}{\pi E} = \frac{1}{15W^2} \left(256 \frac{r_d}{W} + 20 \frac{W}{r_d} \right), \quad (14)$$

$$\bar{E}_{f1} = \frac{E_{f1}}{\pi E} = \frac{W^2}{384} \left[(2\alpha - 1) + 4 \frac{r_d}{W} + 32(2\alpha - 1) \left(\frac{r_d}{W} \right)^2 \right], \quad (15)$$

and for $r_d(t) \leq 4$, we have:

$$\bar{E}_{p2} = \frac{E_{p2}}{\pi E} = \frac{14}{5} \frac{H^2(t)}{r_d^2(t)}, \quad (16)$$

$$\bar{E}_{f2} = \frac{E_{f2}}{\pi E} = \frac{1}{96} [11H^4(t) - (18\alpha + 18)H^3(t) + 36\alpha H^2(t)] r_d^2(t). \quad (17)$$

In the above equations, r_d and W are normalized by the characteristic length L , and H by w_b . For simplicity, we omit the overbar notation for normalized quantities throughout the rest of the analysis.

The total potential energy $V(r_d, H)$, consisting of the elastic energy in the plate and foundation, is given by:

$$V(r_d, H) = \begin{cases} E_{p1} + E_{f1} = V_1(r_d), & \text{if } r_d > \frac{W}{2} = 4, \\ E_{p2} + E_{f2} = V_2(r_d, H), & \text{if } r_d < \frac{W}{2} = 4. \end{cases} \quad (18)$$

As seen from Eqs. (14) and (15), the potential energy in Stage 1, $V_1(r_d)$, consists of three terms: a reciprocal term $\sim 1/r_d$, a linear term $\sim r_d$, and a quadratic term $\sim r_d^2$ (the constant term only affects the energy level but does not change the energy landscape). The reciprocal term originates from the azimuthal bending curvature, which is significant for small r_d but becomes negligible as r_d increases, since the deformation approaches a nearly flat arc and the azimuthal curvature vanishes in the limit of large r_d . The linear term, which has a positive coefficient, corresponds to the surface energy of the phase transition. The coefficient of the quadratic term depends on the parameter α : it is positive for $\alpha > 0.5$ and negative for $\alpha < 0.5$, reflecting the bulk energy gain or loss during the phase transition. These features from the reduced-order model are consistent with qualitative arguments given earlier.

From Eqs. (14)–(17), it follows that the potential energy landscape $V(r_d, H)$ is primarily governed by the bistable foundation parameter α . An illustrative case for $\alpha = 0.48$ is shown in Fig. 4.

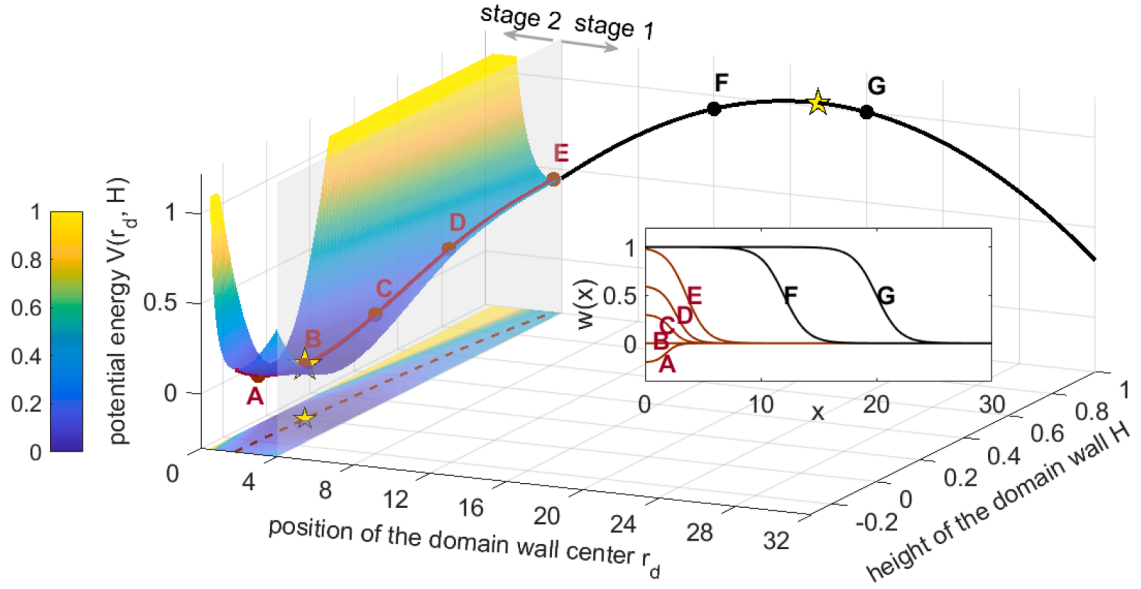


Fig. 4. The potential energy $V(r_d, H)$ of the reduced order model (ROM) with $\alpha = 0.48$. The two stars represent the global maximum in stage 1 (domain wall expanding or shrinking) and the local minimum in stage 2 (domain wall disappearing), respectively. The trajectory FEDCBA is a possible evolution of the domain wall. The insert shows the profiles of the plate with different domain wall positions.

In Stage 1, where the domain wall expands or shrinks, the potential energy depends only on the wall position r_d and exhibits a global maximum at the *critical nucleation size* r_{cr} . If a circular domain wall with $r_{d0} > r_{cr}$ (e.g., point G in Fig. 4) is created and released, the positive driving force $f = -dV/dr_d > 0$ drives it to expand until it reaches the system boundary, restoring the plate to a flat configuration. Conversely, if $r_{d0} < r_{cr}$ (e.g., point F), the negative driving force $f < 0$ causes the wall to contract toward point E. When the evolution reaches point E, the system enters Stage 2, where the potential energy depends on both r_d and H . A local minimum exists at point B, satisfying $\partial V/\partial r_d = 0$ and $\partial V/\partial H = 0$ with a positive definite Hessian. This minimum always occurs at $H = 0$, independent of α , corresponding to the original flat state.

The trajectory in the (r_d, H) space during Stage 2 depends on the incoming velocities $\dot{r}_d(t_E)$ and $\dot{H}(t_E)$. Fig. 4 presents one possible path. In a conservative system, these trajectories form closed loops corresponding to periodic nonlinear oscillations. In dissipative systems, however, energy is gradually lost, and all trajectories ultimately settle into the stable minimum at point B.

3.3. Potential energy for discrete foundation case

Now we consider the foundation in a more discrete case. Our objective is to investigate how the discreteness of the foundation influences the system's potential energy V . As defined previously, with the hexagonal lattice edge size d and the bistable unit stiffness k_{unit} , the effective bistable foundation stiffness per unit area is defined as $k = \frac{4}{3\sqrt{3}} \frac{k_{unit}}{d^2}$ and the characteristic length of the system is defined as $L = \sqrt{\frac{D}{kw_b^2}}$. To approximate the potential energy, when the foundation density is not far away from the continuous limit, we assume that the domain wall has a circular shape and the width of the domain wall is approximately $W = 8L$, the same with the continuous limit. Therefore, the plate deformation still follows the same shape as described by Eq. (6). To quantify the discreteness of the foundation, we have defined the normalized edge size of the hexagon pattern as $\gamma = \frac{d}{W}$. The above assumptions are asymptotic as the results converge to the continuous case when $\gamma \rightarrow 0$. With these assumptions for deformation, the elastic energy due to plate deformation can be integrated in the same way with the continuous case, while the elastic energy in the foundation is obtained by summing the contributions from all bistable units:

$$E_f = \sum_{i=1}^N U_{unit}(w(r_i)) \quad (19)$$

where N is the number of the bistable units in the foundation and r_i is the radial position of the i th unit.

4. Axisymmetric domain walls

In this section, we present the behavior of an axisymmetric domain wall predicted by the ROM in Section 3 and provide quantitative comparison with FEA simulations. We first focus on the continuous limit in Section 4.1 where the domain wall either shrinks or expands, and then we discuss the transition from the continuous limit to discrete foundation case and how the discreteness stabilizes the domain wall in Section 4.2.

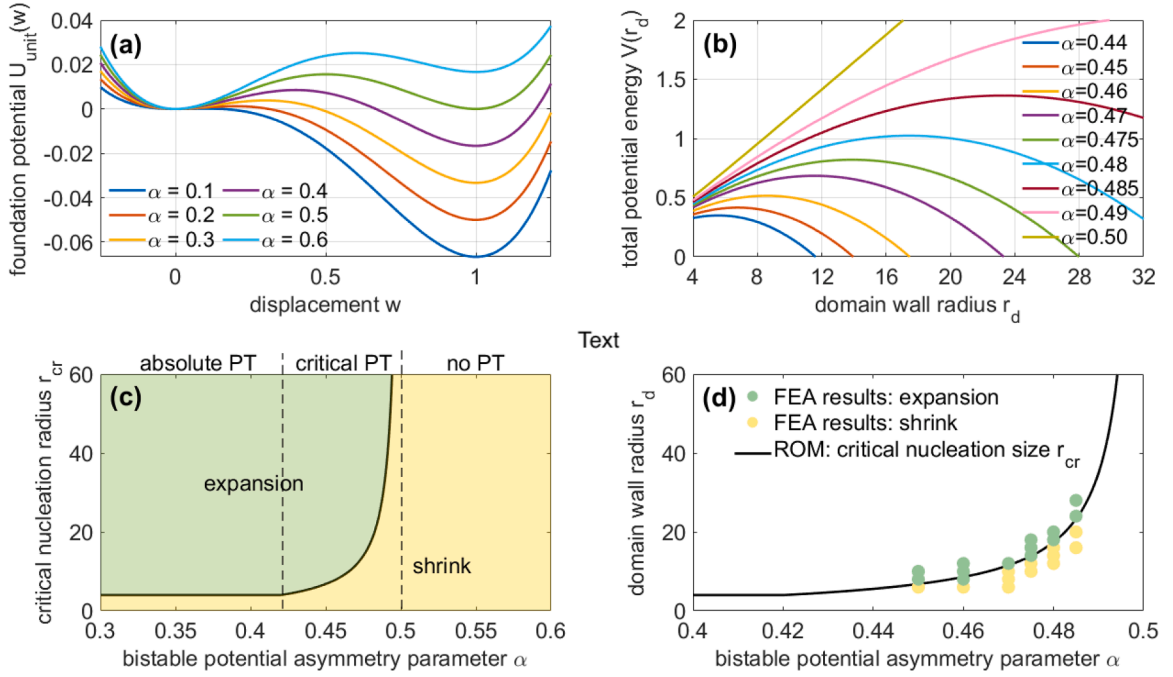


Fig. 5. (a) The potential energy landscape $U_{\text{unit}}(w)$ of a single bistable unit with different α values; (b) the potential energy landscape $V(r_d)$ of the whole system by ROM with different α values; (c) the critical nucleation size r_{cr} as a function of α by the ROM Eq. (21); (d) comparison between the ROM and the FEA simulation results with different α and initial domain wall size r_{d0} . In FEA, $\gamma = \frac{1}{16}$.

4.1. Continuum limit

4.1.1. Critical nucleation size

The critical nucleation size is important as it can be used to determine the minimum size of the actuation area required to trigger a global state change in the system. The critical nucleation size r_{cr} is the position of the global maximum in the potential energy landscape and thus can be determined by:

$$\frac{dV(\alpha, r_d)}{dr_d} \Big|_{r_d=r_{cr}} = 0 \quad (20)$$

Using Eq. (18), the critical nucleation size r_{cr} is the largest root of a cubic equation:

$$\frac{2\alpha - 1}{6} r_{cr}^3 + \left(\frac{W}{96} + \frac{256}{15W^3} \right) r_{cr}^2 - \frac{4}{3W} = 0 \quad (21)$$

In some cases, the largest root lies in Stage 2, or the equation yields only a negative (nonphysical) root. In either situation, $\frac{\partial V(r_d)}{\partial r_d} < 0$ holds for all $r > \frac{W}{2} = 4$, implying that an initial domain wall with $r_d = \frac{W}{2}$ is sufficient to initiate a global phase transition. For these cases we set $r_{cr} = 4$ for simplicity. The dependence of r_{cr} on α is shown in Fig. 5.

As α increases, the critical nucleation size r_{cr} grows significantly. The system can be categorized into three distinct regimes according to the value of α :

1. **Absolute phase transition region:** ($\alpha < \alpha_1 = 0.42$): an initial domain wall with $r_d = W/2$ is sufficient to initiate a global phase transition, and creating a domain wall $r_d = \frac{W}{2}$ only requires a point load. This threshold α_1 can be determined analytically from Eq. (21) when the largest root is equal to $\frac{W}{2}$.
2. **Critical phase transition region:** ($0.42 = \alpha_1 < \alpha < \alpha_2 = 0.5$): there exists a critical domain wall size r_{cr} for which the domain wall expands when the initial domain wall size $r_{d0} > r_{cr}$ and shrinks and finally disappears when the initial domain wall size $r_{d0} < r_{cr}$.
3. **No phase transition region:** ($\alpha \geq 0.5$): the domain wall always shrinks to extinction, regardless of its initial size.

The ROM predictions for r_{cr} are validated against FEA in Fig. 5(d). In FEA, the discreteness parameter is selected as $\gamma = \frac{1}{16}$. For each α value, simulations with various initial radii r_{d0} are performed and large damping coefficients are used to eliminate the effect of initial kinetic energy when the domain wall is released. Green markers denote cases where the domain wall expands, while yellow markers correspond to shrinkage and disappearance. The close agreement between the ROM predictions and the FEA results confirms the accuracy of the reduced-order model in capturing the nucleation threshold.

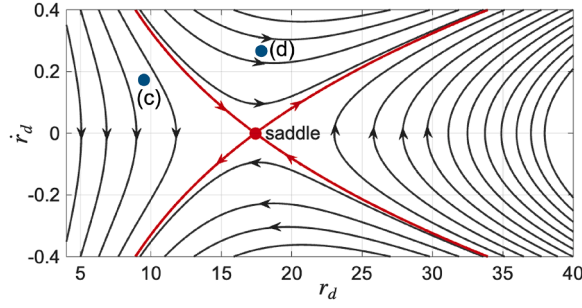


Fig. 6. ROM trajectory of the dynamic evolution of a domain wall in the $(r_d, \dot{r}_d)$ space. The position of the saddle point corresponds to the critical nucleation size r_{cr} . Arrows denote the actual motion directions. Points (c) and (d) denote the initial conditions in Fig. 7(c) and (d). With $\alpha = 0.48$.

4.1.2. Equation of motion: Dynamic domain wall motion

In the previous section, we established the criterion distinguishing between shrinking and expanding domain walls. We now turn to the dynamics, specifically, the rate at which the domain wall moves. To study the dynamic evolution, we formulate the Lagrangian of the reduced-order model in terms of its kinetic and potential energy:

$$L(r_d, \dot{r}_d) = E_k(r_d, \dot{r}_d) - V(r_d) \quad (22)$$

In the ROM stage 1, only $r_d(t)$ evolves and thus the Lagrangian depends only on r_d and its time derivative $\dot{r}_d$. The equation of motion then follows from the Euler-Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}_d} \right) - \frac{\partial L}{\partial r_d} = 0 \quad (23)$$

To obtain the Lagrangian L explicitly, next we derive the kinetic energy E_k . The velocity field of the plate when a domain wall is shrinking or expanding can be obtained from Eq. (7):

$$v(r, t) = \frac{\partial w}{\partial t} = \frac{2w_b}{W} \text{sech}^2 \left(\frac{r - r_d(t)}{W/4} \right) \dot{r}_d \quad (24)$$

and the corresponding kinetic energy is:

$$\begin{aligned} E_{k1} &= \iint \frac{1}{2} \rho h v^2 dA \\ &= \frac{4\pi \rho h w_b^2}{W^2} \dot{r}_d^2 \int_0^\infty \text{sech}^4 \left(\frac{r - r_d}{W/4} \right) r dr \\ &= \frac{4\pi \rho h w_b^2}{3} \frac{r_d}{W} \dot{r}_d^2 \end{aligned} \quad (25)$$

implying an effective mass,

$$m_{\text{eff}} = \frac{8\pi \rho h w_b^2}{3} \frac{r_d}{W}, \quad (26)$$

which scales with the domain wall radius r_d because the kinetic energy is localized near the wall and proportional to its perimeter. The same integration techniques for sech-based functions used in deriving the plate and foundation energies in Eqs. (9) and (10) are also applied in Eq. (25). The complete derivation is included in Appendix B.

Substituting Eq. (25) into the Euler-Lagrange Eq. (23) yields

$$2r_d \ddot{r}_d + \dot{r}_d^2 - \frac{1}{r_d^2} + \frac{W}{8} (2\alpha - 1) r_d + \frac{64}{5W^2} + \frac{W^2}{128} = 0. \quad (27)$$

A first integration of Eq. (27) gives

$$\dot{r}_d^2 = -\frac{W}{16} (2\alpha - 1) r_d - \frac{1}{r_d^2} - \frac{64}{5W^2} - \frac{W^2}{128} + \frac{C}{r_d}, \quad (28)$$

with the integration constant C determined by the initial condition $(r_{d0}, \dot{r}_{d0})$ as

$$C = r_{d0} \left[\frac{W}{16} (2\alpha - 1) r_{d0} + \frac{1}{r_{d0}^2} + \frac{64}{5W^2} + \frac{W^2}{128} + \dot{r}_{d0}^2 \right]. \quad (29)$$

Eq. (28) is equivalent to the energy conservation, relating $\dot{r}_d$ to r_d for any undamped motion. Fig. 6 shows contours of constant C in the $(r_d, \dot{r}_d)$ phase space with parameter $\alpha = 0.48$, each contour representing a trajectory of fixed total energy; arrows indicate the motion direction. The saddle point in Fig. 6 corresponds to the critical nucleation size r_{cr} , i.e., the global maximum of the potential

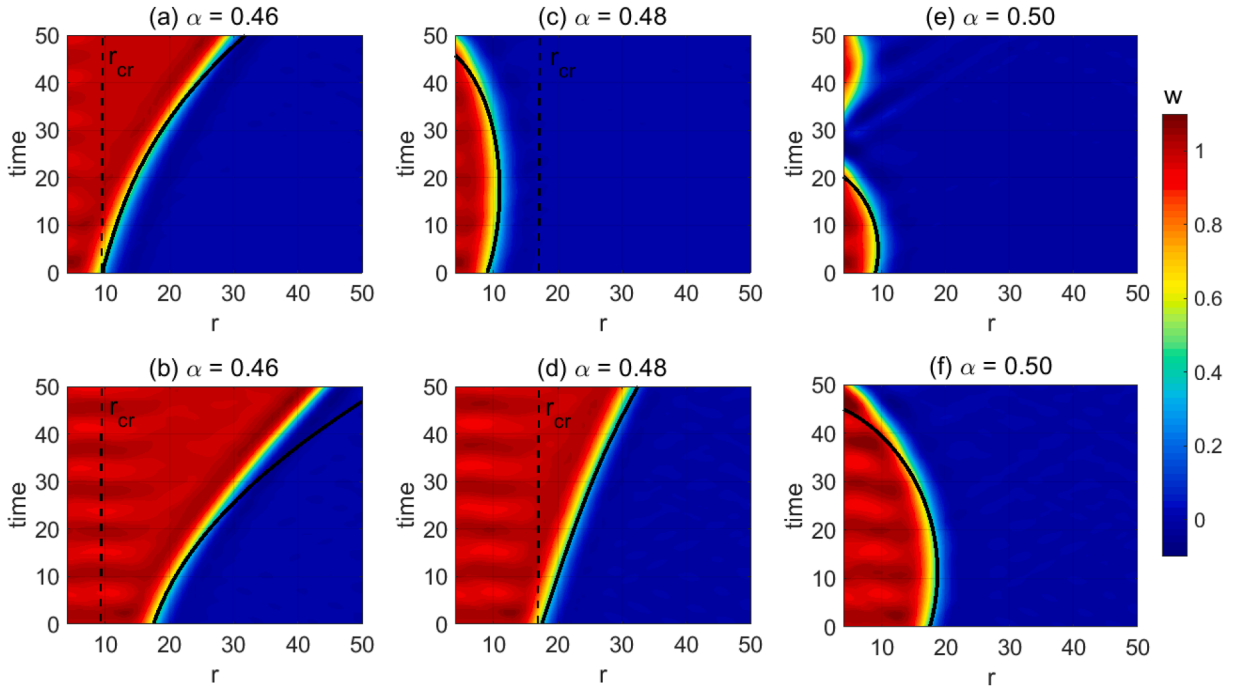


Fig. 7. Domain wall dynamic evolution: displacement w contours obtained from FEA simulations, with the black line showing the domain wall center position predicted by the ROM, Eq. (27). The dashed line denotes the critical nucleation size given by Eq. (21). In FEA, $\gamma = \frac{1}{16}$.

energy landscape. For a wall released from rest, the trajectories show that $r_{d0} < r_{cr}$ leads to shrinkage and $r_{d0} > r_{cr}$ to expansion. The trajectories also show that if the domain wall has a sufficiently large initial velocity ($\dot{r}_{d0} > 0$), it may overcome the potential barrier even when $r_{d0} < r_{cr}$, leading to unbounded expansion. In all cases, the domain wall moving velocity magnitude $|\dot{r}_d|$ increases as r_d moves away from r_{cr} , regardless of whether the wall is shrinking or expanding.

To validate this dynamic ROM, we extracted the initial conditions $r_d(0)$ and $\dot{r}_d(0)$ from FEA simulations and then integrated the ODE Eq. (27) using a 4th-order Runge-Kutta method (ode45 in Matlab) with these initial conditions. Fig. 7 compares the ROM predictions with the FEA results, demonstrating close agreement for both shrinkings and expanding cases. Note that in almost all cases in Fig. 7, the domain wall moves more slowly in the FEA simulations than predicted by the ROM. This discrepancy can be attributed to two main effects. First, after the displacement control is released at the end of the first simulation step (which corresponds to $t = 0$ in Fig. 7), the domain wall not only starts to move inward or outward, but its profile also relaxes from the initially imposed shape toward the profile dynamically favored by the system. This relaxation process generates transient dynamics and radiates small-amplitude waves, thereby transferring part of the kinetic energy associated with domain-wall motion into other dynamic modes. Second, even though the discreteness is relatively small ($\gamma = \frac{1}{16}$) in the FEA simulations, the system is still not truly continuous. As the domain wall propagates through the discrete bistable lattice, it can excite phonons, which radiate energy away from the coherent domain-wall motion. This phonon-radiation behavior has been reported and investigated in detail, for example, in earlier studies of kink motion in discrete bistable one-dimensional lattices (Peyrard and Kruskal, 1984), and more recently in the domain-wall propagation in a beam coupled to a bistable foundation (Jin et al., 2026). In our case, evidence of phonon radiation can be observed from the stripe-like patterns in the contour plots. Since the equation of motion in the ROM, from Eqs. (22) to (27), is derived under a continuous-field assumption, it does not capture phonon radiation and therefore predicts faster domain-wall motion than the FEA.

4.2. Transition from continuum limit to discrete case

4.2.1. ROM prediction

Fig. 8 presents the reduced-order model (ROM) prediction for the effect of the discreteness γ on the potential energy landscape and domain wall stability. In Fig. 8(a), the total potential energy $V(r_d)$ is plotted for a range of bistable potential parameters $\alpha = 0.46, 0.47, 0.48, 0.485, 0.49$ and discreteness parameters $\gamma = 0.1, 0.25, 0.35$. At scales much larger than the lattice size d , the energy landscape is primarily governed by the competition between the bulk energy release and surface energy cost. However, at finer scales (on the order of d), the underlying lattice discreteness γ gives rise to modulations in the potential energy. For small γ (e.g., $\gamma = 0.1$), the energy landscape is nearly identical to the continuous counterpart in Fig. 5. As γ increases, periodic oscillations emerge: the domain wall experiences a periodic variation in total energy as it moves relative to the discrete lattice (Fig. 2), analogous to the Peierls-Nabarro (P-N) effect.

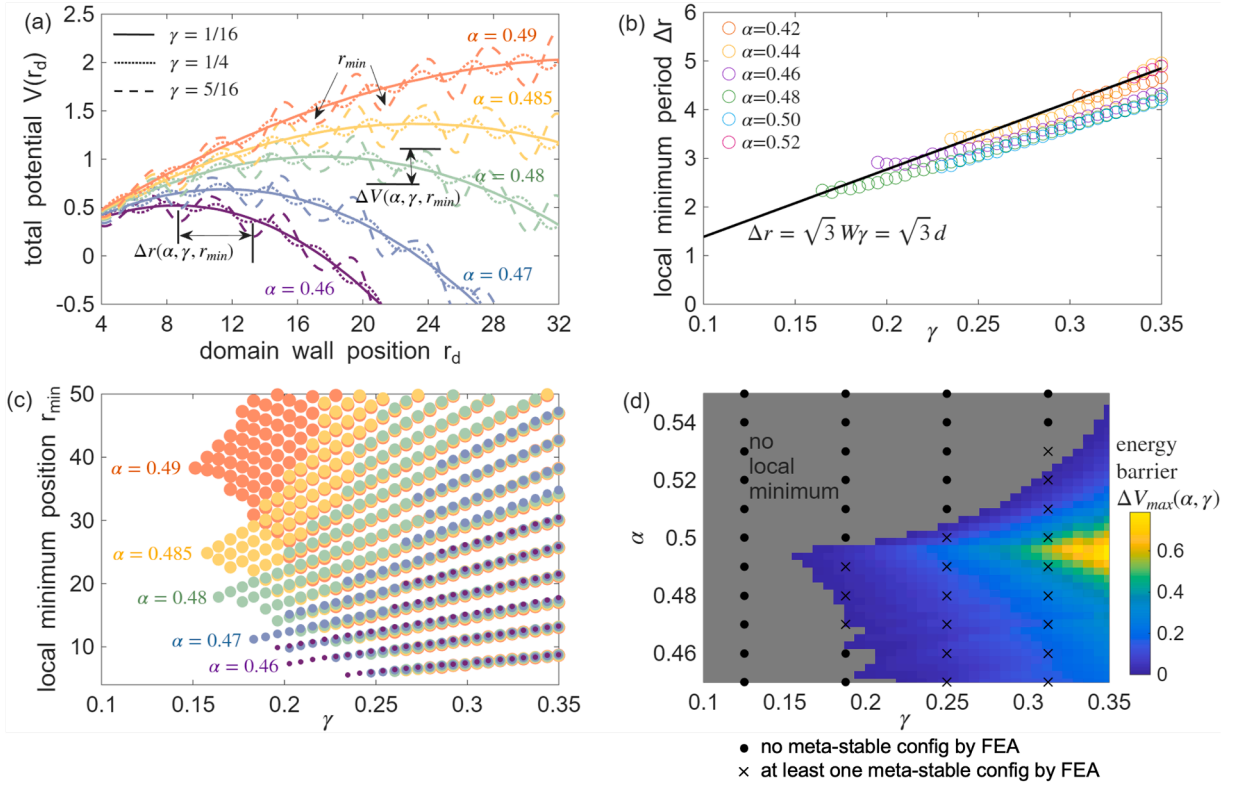


Fig. 8. The ROM prediction for the effect of the discreteness γ on the potential energy landscape and domain wall stability: (a) the energy landscape $V(r_d)$; (b) the spatial period Δr of local minima in energy landscape is proportional to the foundation unit spacing γ ; (c) the spatial distribution of local minima in energy landscape; (d) the maximum energy barrier ΔV_{max} in the potential energy landscape.

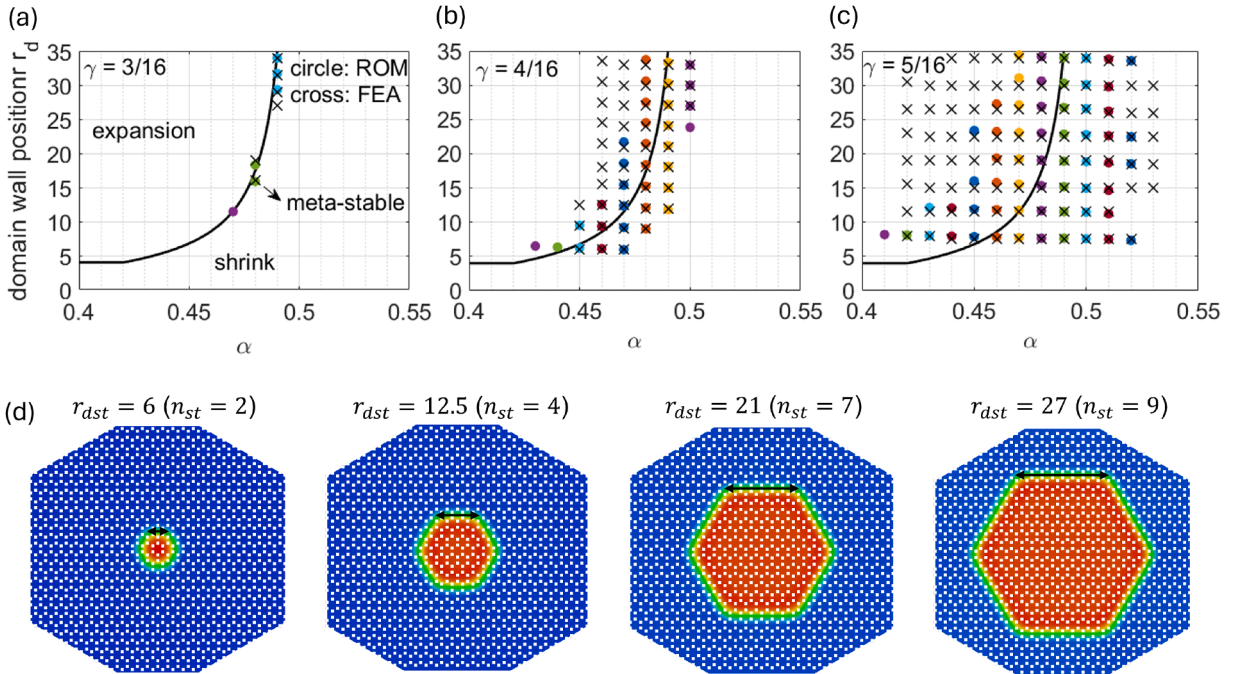


Fig. 9. (a-c) the stable domain wall positions predicted by ROM and FEA simulations for $\gamma = 3/16, 4/16$, and $5/16$, respectively. (d) the stable domain wall configurations in FEA with $\alpha = 0.47$ and $\gamma = 4/16$. The color contours in (d) represent the out-of-plane displacement, and the color bar is identical to that used in Fig. 1.

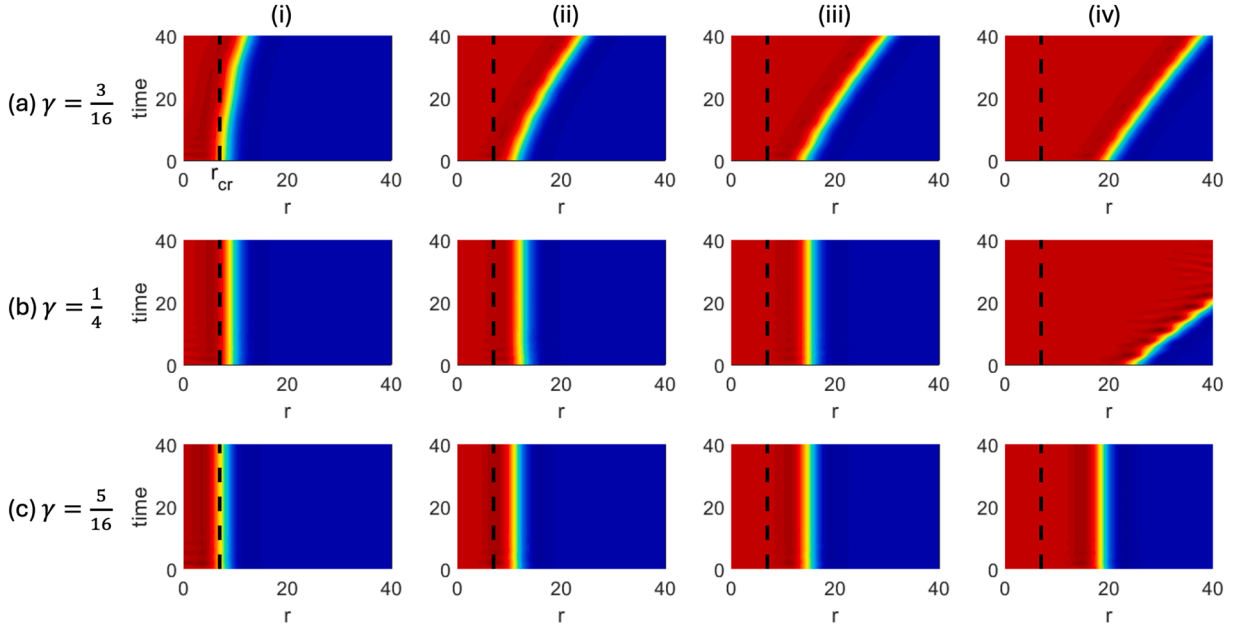


Fig. 10. The domain wall evolution history in FEA for different discreteness values γ . With asymmetry parameter $\alpha = 0.45$. The dashed line represents r_{cr} . The color represents the out-of-plane displacement w , and the color bar is identical to that used in Fig. 1.

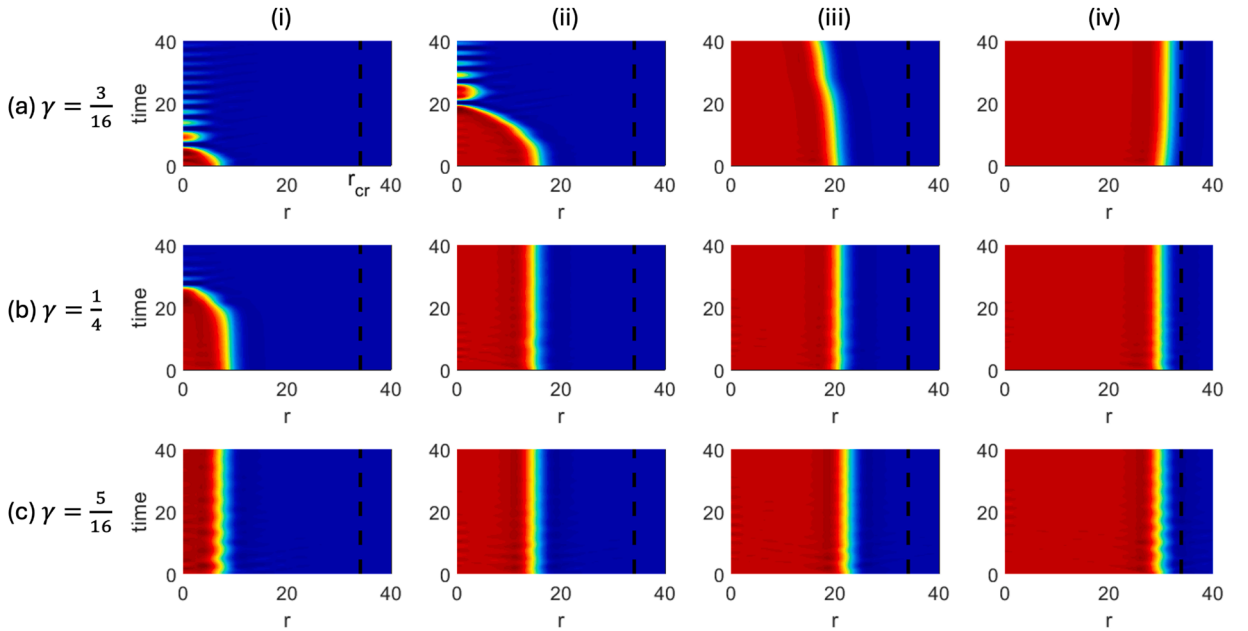


Fig. 11. The domain wall evolution history in FEA for different discreteness values γ . With asymmetry parameter $\alpha = 0.49$. The dashed line represents r_{cr} . The color represents the out-of-plane displacement w , and the color bar is identical to that used in Fig. 1.

The dependence of spatial period Δr of local minima in Fig. 8(a) on parameters α and γ are plotted in Fig. 8(b). The spatial period Δr is largely independent of α and primarily determined by γ . The hexagon foundation lattice has a geometric periodicity of $\sqrt{3}d$ in the radial direction, marked by the black line in Fig. 8(b). The close agreement between the observed energy landscape periodicity Δr and the lattice geometric periodicity confirms that the periodic modulation in the energy landscape is a direct consequence of the underlying discreteness of the foundation.

A domain wall near a local energy minimum r_{min} experiences restoring forces from adjacent energy barriers (local maxima). As a result, small perturbations lead to bounded motion around r_{min} , and dissipation eventually traps the domain wall at such a metastable

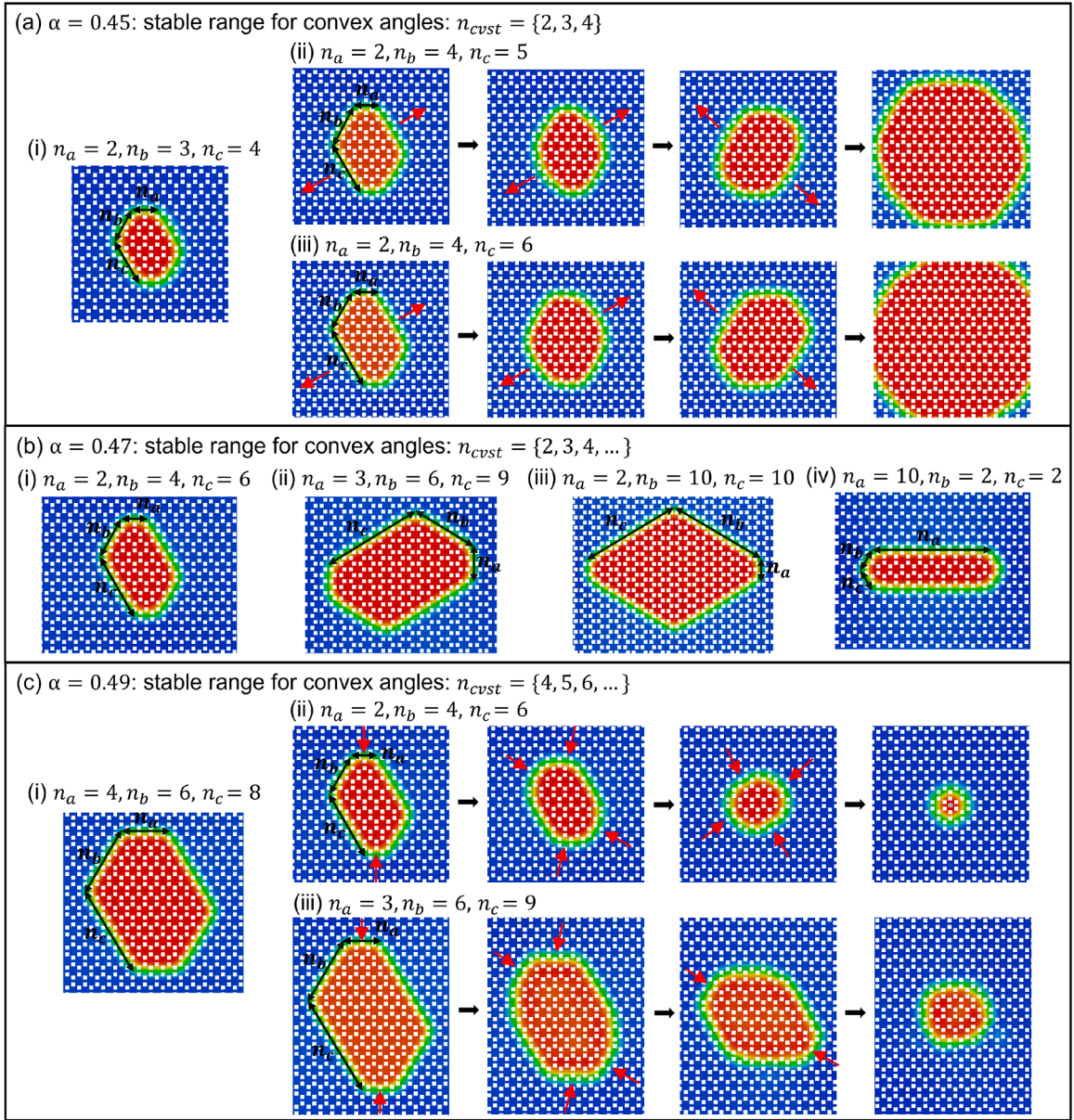


Fig. 12. FEA results for irregular hexagonal domain wall evolution at $\gamma = \frac{1}{4}$: (a) $\alpha = 0.45$: (i) stable configuration; (ii), (iii) unstable and expanding cases; (b) $\alpha = 0.47$: (i)–(iv) all stable configurations; (c) $\alpha = 0.49$: (i) stable; (ii), (iii) unstable and shrinking cases. The color represents the out-of-plane displacement w , and the color bar is identical to that used in Fig. 1.

position. The degree of stability of such a state can be characterized by the energy barrier $\Delta V(\alpha, \gamma, r_{\min})$, defined as the smaller of the two energy differences between a local minimum and its adjacent maxima.

Fig. 8(c) and (d) show the local minimum positions $r_{\min}$ and the maximum energy barriers $\Delta V_{\max}$ across various combinations of α and γ , respectively. For each α , there exists a critical discreteness $\gamma_{\text{cr}}(\alpha)$, beyond which the P-N modulation produces at least one local minimum. Increasing γ beyond γ_{cr} generally leads to more minima and larger energy barriers, indicating stronger domain wall stability. Additionally, three key observations can be made from Fig. 8(c) and (d):

1. For each α , the location of the local-minimum onset consistently aligns with $r_{\text{cr}}(\alpha)$, the location of the global maximum in the corresponding continuum energy curve (see Figs. 5(c) and 8(d)). This alignment occurs because the energy curve is flattest near $r_{\text{cr}}(\alpha)$, making it most sensitive to the P-N modulation. This phenomenon can be more directly observed in Fig. 9(a).

2. For a set of γ and α values, metastable states occur only within a finite range of $r_{\text{dst}}(\gamma, \alpha)$ (Fig. 8(c)). As previously analyzed in Section 2, the periodic P-N energy modulation originates from the bistable units within the domain wall (Fig. 2(b)). Therefore, the magnitude of the P-N modulation scales with r (the perimeter of the domain wall). When r is too small, the oscillation magnitude

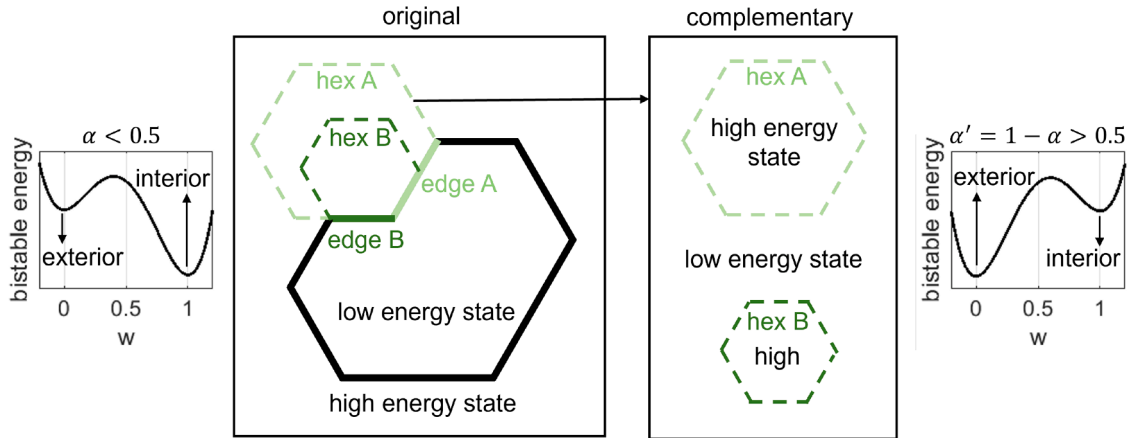


Fig. 13. A domain wall in the shape of an irregular octagon with convex internal angles of 120° and one concave internal angle of 240° , in a system with bistable foundation parameter α . The stability of the concave angle can be inferred from the stability of two complementary regular hexagons A and B with parameter $\alpha' = 1 - \alpha$ that conforms to the concave corner.

is insufficient to generate energy barriers, eliminating local minima; when r is too large, the modulation (which scales with r) is negligible compared to the background energy at a larger scale (which scales as r^2), again suppressing metastability. Hence, metastable domain wall positions are confined to an intermediate range of r_{dst} .

3. In Fig. 8(c) and (d), the smallest value of $\gamma_{\text{cr}}(\alpha)$ occurs as $\alpha \rightarrow 0.5^-$. In the regime close to $\alpha \rightarrow 0.5^-$ the continuous counterpart of the energy curve exhibits a wide flat region near its maximum, making the system particularly sensitive to the P-N effect. As α deviates further from 0.5, a larger discreteness is required to induce the same degree of metastability.

4.2.2. Comparison between ROM and FEA

To validate the ROM in the discrete case, we conducted a series of finite element simulations with different discreteness parameters, γ . The simulation procedure was the same as described previously in Section 2. With sufficiently strong damping, the post-release evolution is strongly dissipative and transient inertial effects are substantially reduced. If the underlying energy landscape contains an accessible local minimum for the given initial condition, the domain wall may undergo damped oscillations and eventually settle into the corresponding metastable configuration. In contrast, if no such local minimum exists, the domain wall continues to evolve by either shrinking or expanding, and no stationary configuration is obtained after sufficiently long simulation time. In this sense, the damping serves to reveal metastable equilibria by dissipating transient energy, while the equilibrium configurations themselves are determined by the underlying energy landscape. To identify the metastable configurations, we repeated the two-step procedure in Section 2 over a wide range of initial radii $r_{d0} \in (0, 35)$ for each value of γ . The sampling interval of r_{d0} was chosen to be the lattice spacing. By scanning these initial conditions and monitoring the long-time evolution after release, we identified the metastable domain-wall sizes reached within the scanned range.

For small values of γ , such as $\gamma = \frac{1}{16}$ and $\frac{2}{16}$, both FEA and ROM show that no metastable domain wall exists for any value of α . The behavior in this regime follows the continuous case, where the domain wall either shrinks or expands according to r_{d0} . Therefore, we focus on the cases with $\gamma > \frac{2}{16}$. Fig. 9(a)–(c) compare FEA results with ROM predictions for $\gamma = \frac{3}{16}, \frac{4}{16}, \frac{5}{16}$, where circles denote local minima position by ROM predictions, crosses represent stable domain wall radius in FEA, and the black line shows the critical radius $r_{\text{cr}}(\alpha)$ from the continuous model (see Section 4.1.1 and Fig. 5). In Fig. 9(a), FEA shows that the onset of stable configurations does occur near $r_{\text{cr}}(\alpha)$, consistent with the ROM prediction. Both FEA and ROM show that for each γ , there exists a range of α values for which the system supports at least one stable domain wall radius. In Figs. 8(d) and 9(a)–(c), this stable range of α broadens with increasing γ , and the ROM predictions agree well with the FEA results. Besides, both FEA and ROM show that the stable range of domain wall radius r_{dst} also broadens with larger γ . The ROM predicts the lower bound of r_{dst} quite well but underestimates its upper bound, indicating that ROM tends to underestimate the domain wall stability for large radii.

The discrepancy between FEA and ROM for large r_d arises from the ROM's assumption of a perfectly circular domain wall. As shown in Fig. 9(d), FEA results show that small domain walls are more similar to circular shapes, aligning well with ROM assumption. However, as r_d increases, the final shape of the domain wall increasingly conforms to the geometry of the hexagonal foundation lattice. For sufficiently large radii, the system evolves into regular hexagonal domain walls, which minimize energy by optimizing alignment with the foundation lattice, giving lower energy than a circular wall of approximately the same size. Consequently, even though the ROM predicts instability for large circular domain walls, the actual system stabilizes by adopting a hexagonal geometry, leading to underestimation of the upper bound of the stable r_{dst} range in the ROM.

The FEA results in Fig. 9(d) also reveal that the stable hexagonal domain walls have side lengths

$$D = \sqrt{3}dn_{\text{st}}$$

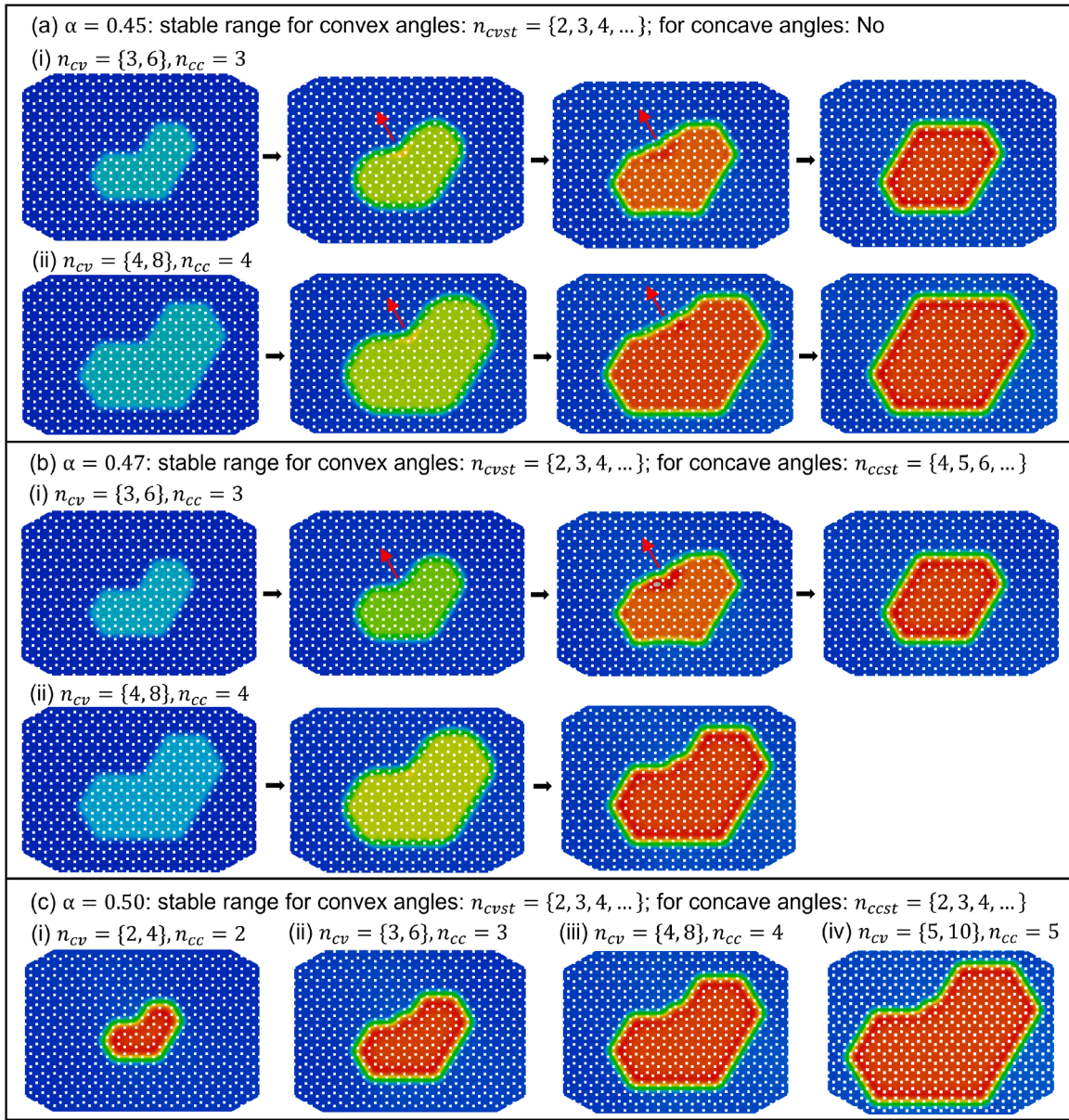


Fig. 14. FEA results of irregular octagonal domain wall evolution at $\gamma = \frac{5}{16}$: (a) $\alpha = 0.45$: (i)(ii) concave angles are unstable and evolve into convex angles; (b) $\alpha = 0.47$: (i) concave angle is unstable and evolves into a convex angle; (ii) stable concave angle; (c) $\alpha = 0.50$: (i)–(iv) all configurations exhibit stable concave angles. The color represents the out-of-plane displacement w , and the color bar is identical to that used in Fig. 1.

where $\sqrt{3}d$ is the spacing between centers of neighboring hexagonal lattices and n_{st} is an integer. Moreover, the orientation of these stable hexagons is aligned with the orientation of the foundation lattice. Based on the stable range of $r_{dst}(\gamma, \alpha)$, one can obtain the corresponding range of stable integers $n_{st}(\gamma, \alpha)$. Describing the stable configurations in terms of n_{st} rather than r_{dst} offers a more convenient geometric parameterization.

Outside the stable r_{dst} range, the system exhibits behaviors analogous to those in the continuous case. Specifically, if the initial size r_{d0} is smaller than the lower bound of r_{dst} , the domain wall shrinks and vanishes. Conversely, if r_{d0} exceeds the upper bound of r_{dst} , the wall expands and fails to stabilize. Figs. 10 and 11 present the dynamic evolution of the domain wall in the FEA for different values of γ and α , illustrating three representative behaviors: shrinking (Fig. 11(ai)–(aiii) and (bi)), stabilization to a metastable state (Figs. 10(bi)–(biii) and (c), and 11(aiv), (bii)–(biv), and (c)), and expansion (Fig. 10(a) and (biv)). Taking the case $\alpha = 0.45$ as an example, the ROM predicts that for $\gamma = 3/16$, no metastable state exists (Fig. 9(a)). Consistent with this prediction, the FEA results in Fig. 10(a) show that all domain walls with initial radius larger than r_{cr} expand. For $\gamma = 4/16$, the ROM predicts two metastable states near r_{cr} (Fig. 9(b)). Correspondingly, the FEA results in Fig. 10(b) show three stable domain-wall sizes near r_{cr} . The stability of the

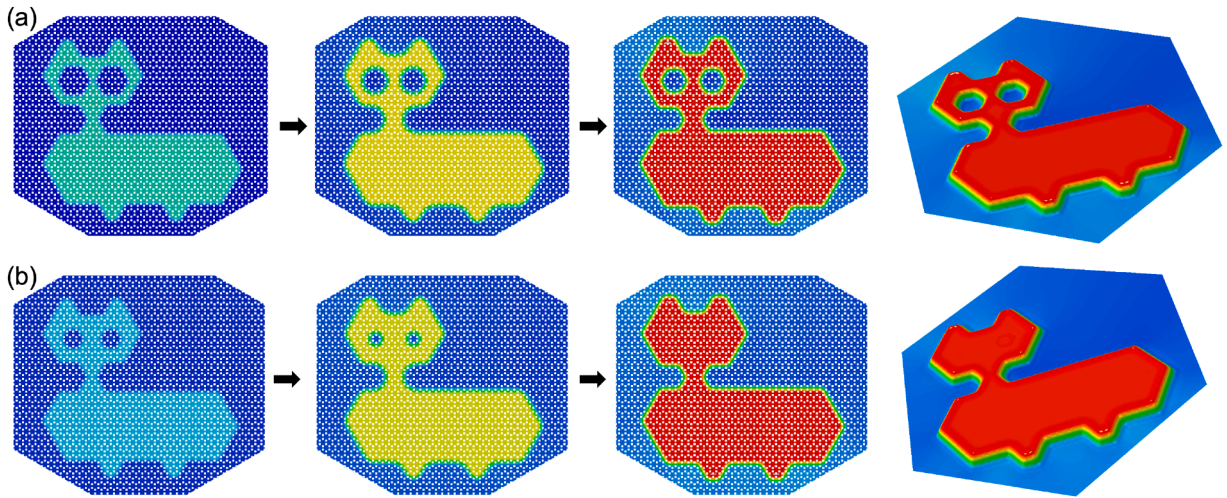


Fig. 15. A cat-shaped geometry is created by one irregular polygon and two hexagonal domain walls. From left to right, the subfigures illustrate the process of applying the prescribed displacement in the green region to generate the initial domain wall and subsequently releasing the displacement constraint. The color represents the out-of-plane displacement w , and the color bar is identical to that used in Fig. 1.

third case in Fig. 10(b) is underestimated by the ROM, for the reason discussed previously. In the fourth case, the larger domain wall expands, which is also consistent with the ROM prediction in Fig. 9(b), as indicated by the negative slope of the energy landscape in Fig. 8(a). For $\gamma = 5/16$, the ROM predicts four metastable states near r_{cr} (Fig. 9(c)), and the FEA results in Fig. 10(c) are in good agreement with this prediction. The results shown in Fig. 11 are similar to those in Fig. 10, but correspond to $\alpha = 0.49$, for which the critical radius r_{cr} is larger.

5. General polygon domain walls

In the continuous limit, for a propagating domain wall the circular shape is the most energy-efficient configuration for front evolution, minimizing the increase in perimeter and thereby the surface energy cost that is required for an increase in area. When the initial nucleation shape deviates from circularity, such as being elliptical or irregular, the system naturally evolves toward a more circular configuration. As discussed in the previous section, a certain degree of foundation discreteness helps pin the axisymmetric (or regular hexagonal) domain wall, preventing it from moving, which makes it possible to have a stable irregular domain wall shape in this system. In this section, we explore the stability of domain walls in irregular shapes and connect the stability of irregular shapes with the axisymmetric (regular hexagonal) case.

5.1. Irregular convex hexagon with internal angles of 120°

As shown in Section 4, for given values of α and γ , there exists a range of integers n_{st} within which regular hexagonal domain walls with side length $D = \sqrt{3}dn_{st}$ are stable. As previously discussed, the prominence of the P-N effect scales with the domain wall perimeter and thus with the side length D . Only within this stable range of n_{st} does the P-N effect sufficiently modulate the energy landscape to enable metastable configurations. Due to the six-fold symmetry of the regular hexagon, the global stability of the entire domain wall is equivalent to the local stability of a segment defined by a single edge of length $D = \sqrt{3}dn_{st}$ and two adjoining interior angles of 120° . This local stability criterion can be extended to assess the global stability of irregular hexagonal domain walls whose interior angles are 120° but side lengths differ.

We define a general irregular hexagon with all the interior angles of 120° and side lengths of $D_i = \sqrt{3}dn_i$, where n_i are integers and $i = 1, \dots, 6$. We expect that if all n_i fall within the stable range of n_{st} for given values of α and γ , then all edges individually satisfy the P-N stability condition, and thus the entire irregular hexagon domain wall remains stable. Conversely, if any of the side lengths fall outside the stable range, the corresponding edges will be energetically unfavorable. Those exceeding the upper bound will tend to expand (move outwards), while those below the lower bound will contract (move inwards). Those edges ultimately lead to the distortion, expansion, or collapse of the irregular hexagonal domain wall.

The above reasoning assumes that the edges of the irregular hexagon behave independently in terms of their stability, and thus the stability of irregular configurations can be inferred from the stability of regular hexagons. This simplification enables a convenient extension of the results obtained in the regular hexagonal cases. However, in reality, adjacent edges and interior angles are geometrically and elastically coupled. The mechanical response of one edge can influence the stresses and deformations in neighboring edges through these geometric constraints. As a result, the stability inference presented above should be regarded as a first-order approximation. To assess its validity, we performed FEA simulations on irregular hexagons with side lengths selected near the boundaries of the stability range obtained from regular hexagons. We generalized the irregular hexagon geometry by introducing three distinct

Table 1

Summary of FEA cases for irregular hexagonal domain walls with convex internal angles 120° . The edge lengths of the hexagon are $D = \sqrt{3}dn$. Discreteness parameter is $\gamma = \frac{1}{4}$.

α	stable range n_{cvt} for 120° angles	n_a	n_b	n_c	inferred stability
0.45	{2, 3, 4}	2	3	4	stable
		2	4	5	edge C expands
		2	4	6	edge C expands
0.47	{2, 3, 4, ... }	2	4	6	stable
		3	6	9	stable
		2	10	10	stable
		10	2	2	stable
0.49	{4, 5, 6, ... }	4	6	8	stable
		2	4	6	edge A shrinks
		3	6	9	edge A shrinks

edge lengths, while preserving the internal angle of 120° at each vertex. The domain wall was constructed as a closed hexagon with the following structure: two edges of length $A = \sqrt{3}dn_a$, two of $B = \sqrt{3}dn_b$, and two of $C = \sqrt{3}dn_c$, where $n_a, n_b, n_c \in \mathbb{Z}^+$ were integer multipliers, and d was the edge length of the foundation unit cell. We selected $\gamma = 1/4$ as with this γ value the lower bounds and upper bounds of n_{st} for different α were distinguished and well separated. The representative cases in our FEA simulation are shown in Table 1. The stability of domain walls in FEA simulations are consistent with the inferred stability, as shown in Fig. 12.

5.2. Irregular polygon with concave internal angles of 240°

To explore a broader class of potentially stable domain wall geometries, we now consider polygons whose internal angles are either 120° or 240° , which are both geometrically compatible with the underlying hexagonal foundation lattice. For edges adjacent to concave angles, we propose the following stability inference. Suppose the foundation inside the polygon is in the low-energy state (i.e., $\alpha < 0.5$), as shown in Fig. 13. For a concave corner (such as the one formed by edges A and B), we construct two conceptual regular hexagons using the same edge lengths as A and B. To maintain the same relative energy distribution on both sides of edge A and B, the interior of each conceptual hexagon must correspond to the high-energy state, while the exterior corresponds to the low-energy state. This inverted energy distribution implies that the conceptual hexagons are governed by a complementary bistable parameter $\alpha' = 1 - \alpha$, leveraging the symmetry of the bistable potential Eq. (1) about $\alpha = 0.5$. We denote these two conceptual regular hexagons as the complementary hexagons of the concave internal angle. If both the complementary hexagons are stable under the complementary bistable parameter α' , we infer that the original edge pair forming the 240° concave angle is also stable. This inference extends the same local-to-global reasoning previously applied in the convex case and relies on the assumption that edge stability is primarily determined by local bistable interactions and geometric compatibility.

Now we apply the above reasoning to our system. From Fig. 9(b), we know that for $\gamma = \frac{1}{4}$, the stable range of α for regular hexagons and therefore for convex 120° internal angles is approximately $[0.45, 0.5]$. The corresponding complementary bistable parameters are $\alpha' = 1 - \alpha \in [0.5, 0.55]$. According to our previous inference, only the value $\alpha' = 0.5$ lies within the known stability range for regular hexagons. Therefore, we predict that only a system with $\alpha = 0.5$ can support stable polygons with both convex and concave angles with this value of γ . For all other values $\alpha' > 0.5$, the corresponding complementary hexagons tend to shrink inward, since the critical domain wall size $r_{\text{cr}} \rightarrow \infty$ for $\alpha > 0.5$. As a result, the original edges adjacent to concave angles tend to expand outward, making the configuration unstable. For a larger discreteness value $\gamma = \frac{5}{16}$, Fig. 9(c) shows that the stable range for convex 120° internal angles broadens to $[0.42, 0.53]$. In this case, both the original parameter α and its complement $\alpha' = 1 - \alpha$ must simultaneously lie within this range $[0.42, 0.53]$ to ensure stability of both convex and concave angles. This leads to the necessary condition for a stable polygon with both convex and concave angles: $0.47 \leq \alpha \leq 0.53$. The full set of sufficient conditions, however, also depends on the edge lengths, as the stability of each segment is determined by its length. To validate this stability inference, we considered octagons with one concave angle of 240° and seven convex angles 120° and conducted FEA simulations. The representative cases are shown in Table 2, along with the predicted stable range of edge lengths for convex and concave angles. The stability of domain walls in FEA simulation are consistent with the inferred stability, as shown in Fig. 14.

5.3. General geometries

Having validated the stability inference for both the convex internal angle 120° and concave angle 240° , now we are able to create more general geometries. Since with $\alpha = 0.50, \gamma = 5/16$, the stable range of n_{st} for 120° and 240° are the same and almost span all the possible edge lengths (Table 2), here we use these parameters in examples. In Fig. 15, three closed domain walls are created to form a cat-like geometry, with the irregular polygon domain wall for the cat's outline and two hexagon domain walls for the cat's eyes. In Fig. 15(b), since the eye domain walls are smaller than the stable range, the eyes evolve inwards and eventually disappear.

Table 2

Summary of FEA cases for irregular octagonal domain wall with convex angles of 120° and one concave angle of 240° . The edge lengths of the hexagon are $D = \sqrt{3}dn$. Discreteness parameter is $\gamma = \frac{5}{16}$.

α	stable range n_{cvst} for 120° convex angles	stable range n_{cst} for 240° concave angles	n_{cv}	n_{cc}	inferred stability
0.45	{2, 3, 4, ...}	No	{3, 6} {4, 8}	3 4	concave angle is unstable concave angle is unstable
0.47	{2, 3, 4, ...}	{4, 5, ..., 8}	{3, 6} {4, 8}	3 4	concave angle is unstable stable
0.50	{2, 3, 4, ...}	{2, 3, 4, ...}	{2, 4} {3, 6} {4, 8} {5, 10}	2 3 4 5	stable stable stable stable

6. Conclusion

This work establishes governing principles on the evolution and stability of closed domain walls in an elastic plate supported by a bistable foundation. Through a reduced-order model (ROM) derived via the collective-coordinate method, supported by finite element simulations, we identify the fundamental mechanisms that determine whether a domain wall expands, shrinks, or arrests at a metastable size.

First, for axisymmetric domain walls, we show that three parameters jointly determine the evolution: the bistable asymmetry α , the discreteness $\gamma = d/W$, and the domain wall radius r_d . In the continuum limit ($\gamma \rightarrow 0$), both FEA and ROM reveal a critical nucleation radius $r_{\text{cr}}(\alpha)$ that separates shrinking and expanding. This behavior arises from the competition between the surface energy, which scales as $O(r_d)$, and the bulk energy difference between the two foundation states, which scales as $O(r_d^2)$. The ROM quantitatively predicts $r_{\text{cr}}(\alpha)$ and its rapid increase as $\alpha \rightarrow 0.5^-$, in good agreement with FEA.

Second, when the discreteness γ exceeds a critical value $\gamma_{\text{cr}}(\alpha)$, the energy landscape is no longer smooth. Instead, periodic Peierls-Nabarro-type (P-N) energy modulations emerge from the discrete foundation, producing local minima in the energy landscape. These minima pin the domain wall and give rise to metastable radii. The ROM correctly predicts (i) that metastability first appears near $r_{\text{cr}}(\alpha)$, the flattest region of the continuum energy landscape; (ii) that $\gamma_{\text{cr}}(\alpha)$ is smallest as $\alpha \rightarrow 0.5^-$; and (iii) that the metastable radius only exists in finite range $r_{\text{dst}}(\alpha, \gamma)$ and this range broadens with increasing γ , arising from the competition between discreteness effects (P-N pinning) and the global energetic drive for phase transition. The ROM accurately captures the lower bound of r_{dst} , and only deviates at large radii, where domain walls in FEA transition from circular to lattice-conforming hexagonal shapes.

Third, we extend the stability analysis to non-axisymmetric geometries. For discrete foundations, the stability of irregular convex polygons with internal angles of 120° is well predicted by the stability range of regular hexagons: if every edge length falls within the stable integer range $n_{\text{st}}(\alpha, \gamma)$, the entire polygon is metastable. For polygons including concave 240° angles, stability can be inferred by introducing complementary regular hexagons governed by the complementary bistable parameter $\alpha' = 1 - \alpha$. FEA confirms that a configuration is stable only when both the convex and concave edges satisfy their corresponding stability criteria. This framework enables the construction of arbitrary metastable shapes by assembling segments with 120° and 240° internal angles.

Overall, this study identifies the critical role of discreteness in creating Peierls-Nabarro-type pinning and establishes design rules linking α , γ , and geometric features to shape stability. In the continuous limit, this study provides quantitative guidance on the minimum actuation area required to trigger a global shape change. In the discrete regime, it identifies the foundation density and system parameters necessary to maintain a desired metastable configuration. Beyond the specific plate-bistable-foundation system investigated here, the energy-landscape framework used in this work can be extended to a broader class of multistable metamaterials composed of elastically coupled bistable units. First, the analysis can be generalized to different lattice types, not limited to the hexagonal honeycomb considered here. As long as the discreteness remains within the regime where a smooth axisymmetric domain-wall profile is a reasonable approximation, the procedure of calculating the potential energy and extracting the P-N-type periodic modulation should remain valid for predicting metastable ranges. Second, the analysis can be extended to a broader class of elastic coupling mechanisms and bistable on-site potentials. The role of the plate in mediating coupling can be interpreted through the biharmonic bending operator, which, when discretized, naturally gives rise to next-nearest-neighbor linear elastic interactions. Other forms of linear or nonlinear elastic coupling, as well as alternative bistable potentials, would modify the quantitative predictions, but the qualitative trends associated with the bistable asymmetry and the discreteness are expected to persist.

This work also has some limitations: (i) We assume that the plate-foundation coupling occurs only through the out-of-plane displacement. Consequently, the analysis focuses on bending-dominated deformations and neglects axial strains and large-deformation effects. This approximation may deviate from experimental conditions when the mechanical connection between the plate and the bistable units is non-ideal or when the plate undergoes significant deformation. (ii) The dynamics of the bistable foundation units are neglected in the present model, where the foundation enters only through its local bistable restoring force. This approximation is most appropriate when the areal mass density of the foundation is small compared with that of the plate. If the foundation units simply follow the local out-of-plane motion of the plate, their main effect would be to increase the effective inertia of the system and thereby slow down the domain-wall motion. However, this limitation mainly affects the transient evolution rather than the static energy landscape. Accordingly, the equilibrium configurations, the energy-based critical nucleation condition, and the existence of

metastable pinned domain walls are still expected to remain qualitatively valid, provided that the bistable energy of the foundation units is well approximated by Eq. (1). (iii) In the present reduced-order modeling framework, the domain wall is approximated by a prescribed profile that is circular in the azimuthal direction and tanh-shaped in the radial direction. This is a first-order approximation in several respects. First, the domain-wall width is assumed to be fixed. However, as observed in kink motion for the beam-bistable system (Decker et al., 2021) and other similar systems (such as $\phi 4$ model, Kevrekidis and Weinstein, 2000), the kink width decreases slightly as the propagation speed increases; thus the present approximation is accurate primarily when the domain-wall motion is slow. In addition, while the circular shape agrees well with FEA for small domain walls, larger domain walls in the discrete system increasingly conform to the hexagonal symmetry of the underlying foundation lattice, which causes the ROM to underestimate the upper bound of the stable domain wall size range. (iv) The stability inference used to generalize axisymmetric results to irregular geometries is a first-order approximation. Additional FEA studies are needed to further validate, refine, or delimit the applicability of this criterion.

CRedit authorship contribution statement

Dengge Jin: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Samuele Ferracin:** Writing – review & editing, Investigation, Formal analysis; **Vincent Tournat:** Writing – review & editing, Supervision, Investigation, Formal analysis, Conceptualization; **Jordan R. Raney:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

None.

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Appendix A. Derivation of the potential energy

When the foundation in a continuous limit where $\gamma = d/W \ll 1$. In stage 1 where the domain wall either shrinks or expands, the elastic energy in the plate (Timoshenko and Woinowsky-Krieger, 1959) is given by

$$\begin{aligned}
 E_{p1} &= \frac{1}{2} D \iint \left[\frac{1}{r^2} \left(\frac{\partial w}{\partial r} \right)^2 + \left(\frac{\partial^2 w}{\partial r^2} \right)^2 \right] dA \\
 &= \pi D w_b^2 \int_0^\infty \left[\frac{4}{W^2} \operatorname{sech}^4 \left(\frac{r-r_d}{W/4} \right) \frac{1}{r} + \frac{16^2}{W^4} \operatorname{sech}^4 \left(\frac{r-r_d}{W/4} \right) \tanh^2 \left(\frac{r-r_d}{W/4} \right) r \right] dr \\
 &= \pi D w_b^2 \int_{-\frac{r_d}{W/4}}^\infty \left[\frac{4}{W^2} \operatorname{sech}^4(\xi) \frac{1}{\frac{W}{4}\xi + r_d} + \frac{16^2}{W^4} \operatorname{sech}^4(\xi) \tanh^2(\xi) \left(\frac{W}{4}\xi + r_d \right) \right] \frac{W}{4} d\xi \\
 &\approx \pi D w_b^2 \int_{-\infty}^\infty \left[\frac{4}{W^2} \operatorname{sech}^4(\xi) \frac{1}{r_d} + \frac{16^2}{W^4} \operatorname{sech}^4(\xi) \tanh^2(\xi) \frac{W}{4} \xi + \frac{16^2}{W^4} \operatorname{sech}^4(\xi) \tanh^2(\xi) r_d \right] \frac{W}{4} d\xi \\
 &= \pi D w_b^2 \int_{-\infty}^\infty \left[\frac{4}{W^2} \operatorname{sech}^4(\xi) \frac{1}{r_d} + \frac{16^2}{W^4} \operatorname{sech}^4(\xi) \tanh^2(\xi) r_d \right] \frac{W}{4} d\xi \tag{A.1}
 \end{aligned}$$

where the Poisson ratio is taken as zero for simplicity without losing the key insights by the ROM. In the second line the change of variable $\frac{r-r_d}{W/4} = \xi$ has been used. In stage 1, $r_d > \frac{W}{2}$ so that $-\frac{r_d}{W/4} < -2$; under this condition, the lower integration limit $-\frac{r_d}{W/4} < -2$ in the third line can be well approximated by $-\infty$ due to the strong localization of sech function, and the first term in the integration can be approximated well by the first term in the fourth line. The second term in the fourth line vanishes as it is odd. Using the identities

$$\int_{-\infty}^\infty \operatorname{sech}^4(\xi) d\xi = \frac{4}{3}, \quad \int_{-\infty}^\infty \operatorname{sech}^4(\xi) \tanh^2(\xi) d\xi = \frac{4}{15} \tag{A.2}$$

yields:

$$E_{p1} = \frac{\pi D w_b^2}{15 W^2} \left(256 \frac{r_d}{W} + 20 \frac{W}{r_d} \right) \tag{A.3}$$

The elastic energy stored in the bistable foundation is

$$\begin{aligned}
 E_{f1} &= \int_0^\infty 2\pi r U(w(r)) dr \\
 &= 2\pi k \int_0^\infty \left[\frac{1}{4} w^4 - \frac{1}{3} (1 + \alpha) w_b w^3 + \frac{1}{2} \alpha w_b^2 w^2 \right] r dr \\
 &= 2\pi k w_b^4 \int_{-\frac{r_d}{W/4}}^\infty \left[\frac{1}{64} \tanh^4 \xi + \left(\frac{1}{48} - \frac{\alpha}{24} \right) \tanh^3 \xi - \frac{1}{32} \tanh^2 \xi + \left(\frac{\alpha}{8} - \frac{1}{16} \right) \tanh \xi + \left(\frac{\alpha}{12} - \frac{5}{192} \right) \right] \\
 &\quad \left(r_d + \frac{W}{4} \xi \right) \frac{W}{4} d\xi,
 \end{aligned} \tag{A.4}$$

where $U(w)$ is the bistable potential Eq. (1) and the ansatz Eq. (7) is used. In stage 1, $r_d > \frac{W}{2}$, so that $-\frac{r_d}{W/4} < -2$, allowing the integrations to be well approximated by

$$\begin{aligned}
 \int_{-z}^\infty (\tanh \xi - 1) \xi d\xi &\approx z^2 - 1, & \int_{-z}^\infty (\tanh^2 \xi - 1) \xi d\xi &= 0 \\
 \int_{-z}^\infty (\tanh^3 \xi - 1) \xi d\xi &= 0 \approx z^2 - 2, & \int_{-z}^\infty (\tanh^4 \xi - 1) \xi d\xi &= 0 \\
 \int_{-z}^\infty (\tanh \xi - 1) d\xi &\approx -2z, & \int_{-z}^\infty (\tanh^2 \xi - 1) d\xi &= -2 \\
 \int_{-z}^\infty (\tanh^3 \xi - 1) d\xi &\approx -2z, & \int_{-z}^\infty (\tanh^4 \xi - 1) d\xi &\approx -\frac{8}{3},
 \end{aligned} \tag{A.5}$$

It can be shown that Eq. (A.4) can be rearranged as a linear combination of the above integrations, and thus it yields to:

$$E_{f1} \approx \frac{\pi k w_b^4 W^2}{384} \left[(2\alpha - 1) + 4 \frac{r_d}{W} + 32(2\alpha - 1) \left(\frac{r_d}{W} \right)^2 \right] \tag{A.6}$$

Next, we consider stage 2 where the domain wall disappears. The elastic energy in the plate in stage 2 follows from Eq. (A.3) with w_b replaced by $H(t)$ and $W(t) = 2r_d(t)$:

$$E_{p2} = \frac{14\pi D}{5} \frac{H^2(t)}{r_d(t)^2} \tag{A.7}$$

The elastic energy in the foundation is:

$$\begin{aligned}
 E_{f2} &= \int_0^\infty 2\pi r U(w(r)) dr \\
 &= 2\pi k \int_0^\infty \left[\frac{1}{4} w^4 - \frac{1}{3} (1 + \alpha) w_b w^3 + \frac{1}{2} \alpha w_b^2 w^2 \right] r dr \\
 &\approx \frac{\pi k w_b^4}{96} \left[11 \left(\frac{H(t)}{w_b} \right)^4 - (18\alpha + 18) \left(\frac{H(t)}{w_b} \right)^3 + 36\alpha \left(\frac{H(t)}{w_b} \right)^2 \right] r_d(t)^2
 \end{aligned} \tag{A.8}$$

which recovers Eq. (A.6) for $H = w_b$ and $r_d = W/2$.

Appendix B. Derivation of the kinetic energy

The kinetic energy in stage 1 can be calculated from the velocity field Eq. (24):

$$\begin{aligned}
 E_{k1} &= \iint \frac{1}{2} \rho h v^2 dA \\
 &= \int_0^\infty \frac{1}{2} \rho h v^2 \cdot 2\pi r dr \\
 &= \frac{4\pi \rho h w_b^2}{W^2} r_d^2 \int_0^\infty \operatorname{sech}^4 \left(\frac{r - r_d}{W/4} \right) r dr
 \end{aligned} \tag{B.1}$$

For this integration, a change of variable $\frac{r - r_d}{W/4} = \xi$ yields

$$\begin{aligned}
 \int_0^\infty \operatorname{sech}^4 \left(\frac{r - r_d}{W/4} \right) r dr &= \int_{-\frac{r_d}{W/4}}^\infty \operatorname{sech}^4(\xi) \left(\frac{W}{4} \xi + r_d \right) \frac{W}{4} d\xi \\
 &= \frac{W^2}{16} \int_{-\frac{r_d}{W/4}}^\infty \operatorname{sech}^4(\xi) \xi d\xi + \frac{W}{4} r_d \int_{-\frac{r_d}{W/4}}^\infty \operatorname{sech}^4(\xi) d\xi
 \end{aligned} \tag{B.2}$$

Due to $-\frac{r_d}{W/4} < -2$, the lower integration limit can be well approximated by $-\infty$ due to the localized property of sech function. The first term of the integration vanishes since $\text{sech}^4(\xi)\xi$ is odd, and the second term evaluates to

$$\int_{-\infty}^{\infty} \text{sech}^4(\xi) d\xi = \frac{4}{3} \quad (\text{B.3})$$

Therefore, the kinetic energy in Eq. (B.1) becomes

$$E_{k1} = \frac{4\pi\rho h w_b^2}{3} \frac{r_d}{W} \dot{r}_d^2 \quad (\text{B.4})$$

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